# Languages and Computation

## Discrete Math, Spring 2026

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## Graph Theory

- Graphs & digraphs
- Paths & connectivity
- Trees & spanning trees
- Bipartite graphs
- Matchings & Hall's theorem
- Planarity & coloring
- Network flows

## Languages & Computation

- Alphabets & formal languages
- Regular expressions
- Finite automata (DFA, NFA)
- Pumping lemma
- Context-free grammars
- Pushdown automata
- Turing machines
- Decidability & complexity

## Combinatorics & Recurrences

- Counting principles
- Permutations & combinations
- Inclusion–exclusion
- Partitions & Stirling numbers
- Generating functions
- Recurrence relations
- Asymptotic analysis

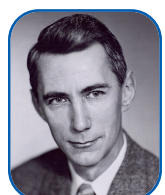
# Formal Languages

*"The limits of my language mean the limits of my world."*

***— Ludwig Wittgenstein***



Noam Chomsky



Claude Shannon

# Why Formal Languages?

Every non-trivial computation involves *processing structured input*:

- Compilers read *programs* (strings over some alphabet).
- Databases evaluate *queries* (strings with specific syntax).
- Network protocols parse *packets* (sequences of bytes).
- Bioinformatics analyzes *DNA* (strings over $\{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\}$).

The theory of formal languages gives us a *mathematical framework* for answering:

- What kinds of patterns can we *describe*?
- What kinds of patterns can we *recognize* (and how efficiently)?
- What are the *fundamental limits* of computation?

**Key insight:** The study of formal languages is not just about strings — it is the foundation of the *theory of computation*.

Every computational problem can be phrased as: "Given a string $w$, does $w$ belong to language $L$?"

# Basic Terminology

**Definition 1**: *Alphabet* $\Sigma$ is a finite non-empty set of symbols.

*Examples*: $\Sigma_1 = \{\mathrm{a}, \mathrm{b}, \mathrm{c}\}$, $\Sigma_2 = \{0, 1\}$, $\Sigma_3 = \{$🦀, 🐈, 🦭, 🦁$\}$.

**Definition 2**: A *word*, or a *string*, over $\Sigma$ is a *finite* sequence of symbols from $\Sigma$.

*Examples*: “abacaba”, “10110001”, “i am a word”, “” (empty word $\varepsilon$).

**Definition 3**: The set of *all* finite words over the alphabet $\Sigma$ is called the *Kleene star*, $\Sigma^* = \bigcup_{k=0}^{\infty} \Sigma^k$.

**Definition 4**: A *formal language* $L \subseteq \Sigma^*$ is a set of finite words over a finite alphabet.

*Examples*: $L_1 = \{0, 001, 0001, \ldots\}$, $L_2 = \{\mathrm{a}, \mathrm{aba}, \mathrm{ababa}, \mathrm{abababa}, \ldots\}$, $L_3 = \emptyset$, $L_4 = \{\varepsilon, \mathrm{ricercar}\}$.

# Operations on Languages

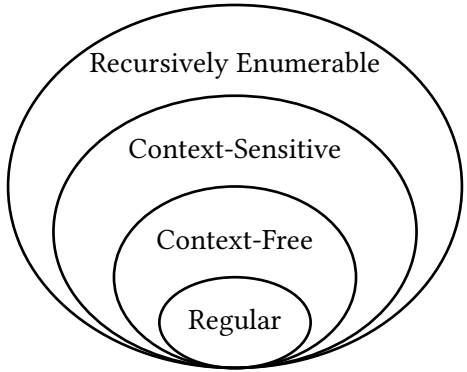
- A formal language, $L \subseteq \Sigma^*$, can be defined by:
  - an *enumeration* of words, *e.g.* $L = \{w_1, w_2, ..., w_n\}$
  - a *regular expression*, *e.g.* $L \triangleq \texttt{01*}$
  - a *formal grammar*, *e.g.* $L \cong G$
- *Set-theoretic* operations:
  - $L_1 \cup L_2 = \{w \mid w \in L_1 \vee w \in L_2\}$, the *union* of $L_1$ and $L_2$
  - $\overline{L} = \{w \mid w \notin L\} = \Sigma^* \setminus L$, the *complement* of $L$
  - $|L|$ is the *cardinality* of $L$
- *Concatenation*:
  - $L_1 \cdot L_2 = \{ab \mid a \in L_1, b \in L_2\}$, where $ab$ is the concatenation of words $a$ and $b$.
  - $L^k = \underbrace{L \cdot ... \cdot L}_{k \text{ times}} = \{w_1 w_2 ... w_k \mid w_i \in L\}$
  - $L^0 = \{\varepsilon\}$
- *Kleene star*: $L^* = \bigcup_{k=0}^{\infty} L^k$

# Chomsky Hierarchy

Formal languages are classified by *Chomsky hierarchy* — a nested family of increasingly powerful language classes, each recognized by a correspondingly more powerful machine.

| Type | Class | Machine | Example |
|---|---|---|---|
| 3 | Regular | Finite Automata | $\{a^n \mid n \geq 0\}$ |
| 2 | Context-Free | Pushdown Automata | $\{a^n b^n \mid n \geq 0\}$ |
| 1 | Context-Sensitive | Linear-Bounded TMs | $\{a^n b^n c^n \mid n \geq 0\}$ |
| 0 | Recursively Enumerable | Turing Machines | $\{\langle M, w\rangle \mid M \text{ halts on } w\}$ |

# Chomsky Hierarchy [2]




Each level of the hierarchy represents a *trade-off* between *expressive power* and *decidability*.

More expressive languages come at the cost of harder (or impossible) algorithmic questions.

# Decision Problems as Languages

Any *decision problem* — a question with a "yes" or "no" answer — can be reformulated as a *language membership test*. The language encodes all inputs for which the answer is "yes".

**Definition 5**: A *decision problem* is a question with a "yes" or "no" answer depending on the input. Formally, the set of inputs for which the answer is "yes" is a language $L \subseteq \Sigma^*$.

Deciding the problem means deciding membership in the language $L$.

**Satisfiability (SAT):** Given a Boolean formula $\varphi$, is it satisfiable?

$$\text{SAT} = \{\varphi \mid \varphi \text{ is a satisfiable Boolean formula}\}$$

**Validity (VALID):** Given a Boolean formula $\varphi$, is it a tautology?

$$\text{VALID} = \{\varphi \mid \varphi \text{ is a valid (universally true) formula}\}$$

**Halting Problem (HALT):** Given a TM $M$ and input $w$, does $M$ halt on $w$?

$$\text{HALT} = \{\langle M, w\rangle \mid \text{TM } M \text{ halts on input } w\}$$

# Decision Problems as Languages [2]

Asking "is $w$ in $L$?" and asking "does the algorithm say yes on input $w$?" are *the same question*.

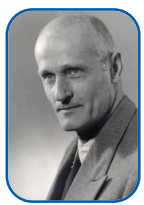
This lets us use the theory of *formal languages* to study the limits of *computation*.

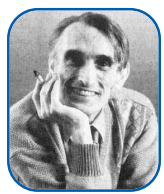
# Regular Languages

*"A language that doesn't affect the way you think about programming, is not worth knowing."*

***— Alan Perlis***


Ken Thompson


Stephen Kleene


Marcel-Paul Schützenberger


Alfred Aho


John Hopcroft


Jeffrey Ullman

# Regular Languages

**Definition 6**: A class of regular languages REG is defined inductively:

- $\mathrm{Reg}_0 = \{\emptyset, \{\varepsilon\}\} \cup \{\{a\} \mid a \in \Sigma\}$, the *empty* and *singleton* languages.
- $\mathrm{Reg}_{i+1} = \mathrm{Reg}_i \cup \{A \cup B \mid A, B \in \mathrm{Reg}_i\} \cup \{A \cdot B \mid A, B \in \mathrm{Reg}_i\} \cup \{A^* \mid A \in \mathrm{Reg}_i\}$, the inductively extended $(i+1)$-th *generation* of regular languages.
- $\mathrm{REG} = \bigcup_{k=0}^{\infty} \mathrm{Reg}_k$, the *class* of all regular languages.

**Theorem 1**: REG is closed under union, concatenation, and Kleene star operations.

**Proof**: Let $A \in \mathrm{Reg}_i$, $B \in \mathrm{Reg}_j$.

- $(A \cup B) \in \left(\mathrm{Reg}_i \cup \mathrm{Reg}_j\right) \in \mathrm{Reg}_{\max(i,j)+1} \subseteq \mathrm{REG}$
- $(A \cdot B) \in \left(\mathrm{Reg}_i \cdot \mathrm{Reg}_j\right) \in \mathrm{Reg}_{\max(i,j)+1} \subseteq \mathrm{REG}$
- $A^* \in \mathrm{Reg}_{i+1} \subseteq \mathrm{REG}$

□

# Regular Expressions

Regular languages can be composed from "smaller" regular languages.

- *Atomic* regular expressions:
  - $\emptyset$, an empty language
  - $\varepsilon$, a singleton language consisting of a single $\varepsilon$ word
  - `a`, a singleton language consisting of a single 1-letter word $a$, for each $a \in \Sigma$
- *Compound* regular expressions:
  - $R_1 R_2$, the concatenation of $R_1$ and $R_2$
  - $R_1 \mid R_2$, the union of $R_1$ and $R_2$
  - $R^* = RRR\ldots$, the Kleene star of $R$
  - $(R)$, just a bracketed expression
  - Operator precedence: `ab*c|d` $\triangleq$ `((a (b*)) c) | d`

# Summary of Regular Expressions

| Language | Expression | Description |
|---|---|---|
| $\emptyset$ | | Empty language |
| $\{\varepsilon\}$ | $\varepsilon$ | Language with a single empty word |
| $\{\mathrm{a}\}$ | $\mathtt{a}$ | Singleton language with a literal character “a” |
| $A$ | $\alpha$ | Language $A$ denoted by regex $\alpha$ |
| $B$ | $\beta$ | Language $B$ denoted by regex $\beta$ |
| $A \cup B$ | $\alpha \mid \beta$ | Union of languages $A$ and $B$ |
| $A \cdot B$ | $\alpha\beta$ | Concatenation of languages $A$ and $B$ |
| $A^*$ | $\alpha^*$ | Kleene star of language $A$ |
| $A^+$ | $\alpha^+$ | Kleene plus of language $A$ |

# Reading Regular Expressions

*Example*: `(a|bc)*` $= \{\varepsilon, \text{a}, \text{aa}, \text{aaa}, \ldots, \text{bc}, \text{bcbc}, \text{bcbcbc}, \ldots, \text{abc}, \text{bca}, \text{abca}, \text{abcbc}, \text{bcabc}, \ldots\}$

*Example*: `0(10)*1` $= \{01, 0101, 010101, \ldots\}$

See also: PCRE

**Key insight:** Regular expressions describe *exactly* the regular languages. This is not obvious — it takes Kleene's Theorem (stated later) to prove it.

# Finite Automata

*"We may hope that machines will eventually compete with men in all purely intellectual fields."*

***— Alan Turing***



Warren McCulloch



Walter Pitts



Victor Glushkov



Janusz Brzozowski



Michael Rabin

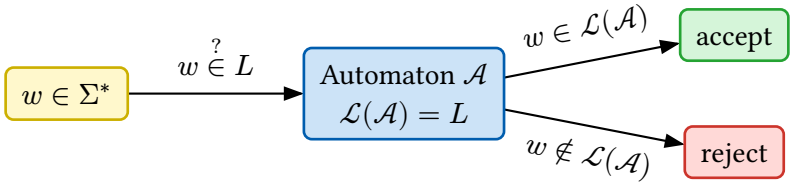


Dana Scott

# Recognizers and Transducers

There are two main types of finite-state *machines*:

1. *Acceptors* (or *recognizers*), automata that produce a binary *yes/no answer*, indicating whether or not the received input word $w \in \Sigma^*$ is *accepted*, *i.e.*, belongs to the language $L$ recognized by the automaton.




2. *Transducers*, machines that produce an output action *for each* symbol of an input.
   - Moore machines (1956)
   - Mealy machines (1955)

In this course, we focus on *acceptors*.

# Deterministic Finite Automata

**Definition 7**: A *Deterministic Finite Automaton* (DFA) is a 5-tuple $\mathcal{A} = (Q, \Sigma, \delta, q_0, F)$ where:

- $Q$ is a *finite* set of states,
- $\Sigma$ is an *alphabet* (finite set of input symbols),
- $\delta : Q \times \Sigma \to Q$ is a *transition function*,
- $q_0 \in Q$ is the *start* state,
- $F \subseteq Q$ is a set of *accepting* (*final*) states.

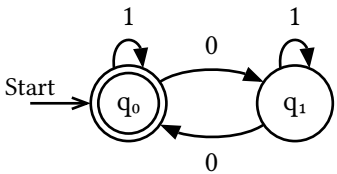
A DFA is a machine that reads an input string *one symbol at a time*, transitioning between states. After reading the entire input, the machine either *accepts* or *rejects* based on whether it ended in an accepting state.

**Note**: *Deterministic* means: at every state, for every symbol, there is *exactly one* transition.

# Example of a DFA

*Example*: Automaton $\mathcal{A}$ recognizing strings with an even number of 0s, $\mathcal{L}(\mathcal{A}) = \{w \in \{0,1\}^* \mid w \text{ has even number of 0s}\}$.

| | 0 | 1 |
|---|---|---|
| q0 | q1 | q0 |
| q1 | q0 | q1 |




Here, $q_0$ is the *start* (denoted by an arrow) and also the *accepting* (denoted by double circle) state.

On input 1, the machine stays.
On input 0, the machine toggles.

# Computation in a DFA

**Definition 8**: A process of *computation* by a finite-state machine $\mathcal{A}$ is a finite sequence of *configurations*, or *snapshots*. A set of all possible configurations is denoted $\mathrm{SNAP} = Q \times \Sigma^*$.

**Definition 9**: A *reachability relation* $\vdash$ is a binary relation over configurations:

$$\langle q, \alpha \rangle \vdash \langle r, \beta \rangle \quad \text{iff} \quad \begin{cases} \alpha = c\beta & \text{where } c \in \Sigma \\ r = \delta(q, c) \end{cases}$$

- $c_1 \vdash c_2$ means "configuration $c_2$ is reachable in *one step* from $c_1$".
- $\vdash^*$, the reflexive-transitive closure of $\vdash$, denotes "reachable in *any* number of steps".

**Note**: A configuration is a complete snapshot of the computation: once the current state and the remaining input are fixed, the future behavior is determined.

# Automata Languages

**Definition 10**: A word $w \in \Sigma^*$ is *accepted* by an automaton $\mathcal{A}$ if the computation, starting in the initial configuration at state $q_0$ with input $w$, *can reach the final configuration* $\langle f, \varepsilon \rangle$, where $f \in F$ is any accepting state, and $\varepsilon$ denotes that the input has been fully consumed.

Formally, $\mathcal{A}$ *accepts* $w \in \Sigma^*$ if $\langle q_0, w \rangle \vdash^* \langle f, \varepsilon \rangle$ for some $f \in F$.

**Definition 11**: The language *recognized* by an automaton $\mathcal{A}$ is a set of all words accepted by $\mathcal{A}$.

$$\mathcal{L}(\mathcal{A}) = \{w \in \Sigma^* \mid \langle q_0, w \rangle \vdash^* \langle f, \varepsilon \rangle \text{ where } f \in F\}$$

**Definition 12**: The class of *automaton languages* recognized by DFAs is denoted AUT.

$$\text{AUT} = \{X \mid \exists \mathcal{A} \text{ such that } \mathcal{L}(\mathcal{A}) = X\}$$

# DFA Exercises

For each language below (over the alphabet $\Sigma = \{0, 1\}$), draw a DFA recognizing it:

1. $L_1 = \{101, 110\}$
2. $L_2 = \Sigma^* \setminus \{101, 110\}$
3. $L_3 = \{w \mid w \text{ starts and ends with the same bit}\}$
4. $L_4 = \{110\}^* = \{\varepsilon, 110, 110110, 110110110, ...\}$
5. $L_5 = \{w \mid w \text{ contains } 110 \text{ as a substring}\}$

# Non-deterministic Automata

# Non-deterministic Finite Automata

**Definition 13**: A *Non-deterministic Finite Automaton* (NFA) is a 5-tuple $\mathcal{A} = (Q, \Sigma, \delta, q_0, F)$, where

- $Q$ is a *finite* set of states,
- $\Sigma$ is an *alphabet* (finite set of input symbols),
- $\delta : Q \times \Sigma \to \mathcal{P}(Q)$ is a *transition function*,
- $q_0 \in Q$ is an *initial* (*start*) state,
- $F \subseteq Q$ is a set of *accepting* (*final*) states.

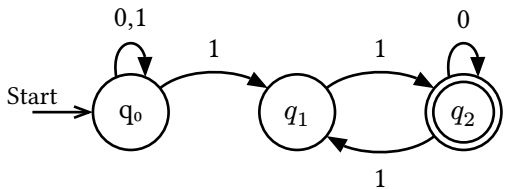
The key difference from a DFA: the transition function returns a *set* of possible next states.

**Note**: $\delta : (q, c) \mapsto \underbrace{\{q^{(1)}, ..., q^{(n)}\}}_{\text{non-determinism}}$

# Example of an NFA

Consider the following NFA recognizing $\mathcal{L}(\mathcal{A}) = \Sigma^*(110^*)^+$ — the strings containing at least one `11` followed by any number of `0`s:

| | 0 | 1 |
|---|---|---|
| q0 | q0 | q0, q1 |
| q1 | | q2 |
| q2 | q2 | q1 |




This NFA has *two* transitions from $q_0$ by the symbol 1: it can go to $q_0$ or to $q_1$.

If an NFA needs to make a non-existent transition (for example, at $q_1$ on input 0), that computation branch simply terminates and rejects.

# Determinism and Non-determinism

**Definition 14**: A model of computation is *deterministic* if at every point in the computation, there is exactly *one choice* that can be made.

**Note**: The machine accepts if *that* series of choices leads to an accepting state.

**Definition 15**: A model of computation is *non-deterministic* if the computing machine may have *multiple decisions* that it can make at one point.

**Note**: The machine accepts if *any* series of choices leads to an accepting state.

**Key insight:** Non-determinism is not about randomness. A non-deterministic machine accepts if *there exists* at least one accepting path — it doesn't matter how unlikely that path might be.

# Intuitions on Non-determinism

There are three useful ways to think about non-determinism:

**1. Tree Computation**

At each *decision point*, the automaton *clones* itself for each possible decision. The series of choices forms a directed, rooted *tree*. If *any* leaf is accepting, the machine accepts.

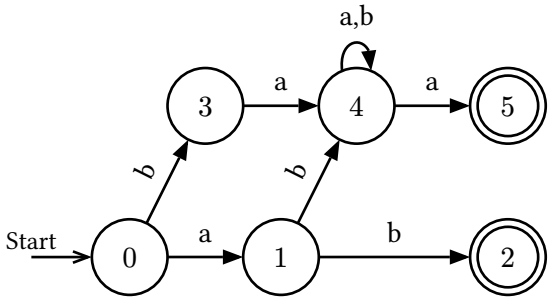
**2. Perfect Guessing**

A non-deterministic machine can be imagined as making an idealized correct guess whenever an accepting branch exists. This is only an intuition; actual machines must simulate the branching explicitly.

**3. Massive Parallelism**

An NFA can be thought of as a machine that tries *all possibilities in parallel*, using an unlimited number of "processors". Each symbol read causes a transition on every currently active state.

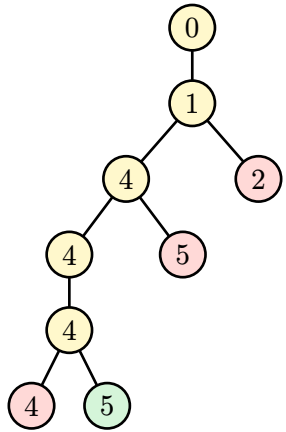
# Tree Computation




**Input:**
$w = \mathrm{a\ b\ a\ b\ a}$

**How to read this:** each time there are multiple outgoing transitions, the NFA branches into multiple active computations.

# Accepting Branches




- At each *decision point*, the automaton *clones* itself for each possible decision.
- At the end, if *any* active accepting (green) states remain, we *accept*.

# NFA Computation Model

Reachability relation for NFA is very similar to DFA's:

$$\langle q, x\rangle \vdash_{\text{DFA}} \langle r, y\rangle \quad \text{iff} \quad \begin{cases} x = cy & \text{where } c \in \Sigma \\ r = \delta(q, c) \end{cases}$$

$$\langle q, x\rangle \vdash_{\text{NFA}} \langle r, y\rangle \quad \text{iff} \quad \begin{cases} x = cy & \text{where } c \in \Sigma \\ r \in \delta(q, c) \end{cases}$$

**Definition 16**: An NFA *accepts* a word $w \in \Sigma^*$ iff $\langle q_0, w\rangle \vdash^* \langle f, \varepsilon\rangle$ for some $f \in F$.

**Definition 17**: A language *recognized* by an NFA is a set of all words accepted by the NFA.

$$\mathcal{L}(\mathcal{A}) = \{w \in \Sigma^* \mid \langle q_0, w\rangle \vdash^* \langle f, \varepsilon\rangle, f \in F\}$$

# DFAs and NFAs

| **Deterministic (DFA)** | **Non-Deterministic (NFA)** |
|---|---|
| Single transition per symbol | Multiple possible transitions |
| No $\varepsilon$-transitions | May have $\varepsilon$-transitions |
| Unique computation path | Multiple parallel paths |
| $\delta : Q \times \Sigma \to Q$ | $\delta : Q \times \Sigma \to \mathcal{P}(Q)$ |
| Easy to simulate | Hard to simulate directly |
| May need exponentially many states | Can be exponentially more concise |

**Why NFAs?**

Despite having the same expressive power as DFAs, NFAs are often *exponentially more concise*.

They are also the natural output of many constructions (*e.g.*, Thompson's construction).

# $\varepsilon$-NFA

**Definition 18**: An *$\varepsilon$-NFA* is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ with the same components as an NFA, but with a modified transition function:

$$\delta : Q \times (\Sigma \cup \{\varepsilon\}) \to \mathcal{P}(Q)$$

From any state the machine may follow *$\varepsilon$-transitions* without consuming any input symbol.

**Definition 19**: *Epsilon closure* of a state $q$, denoted $E(q)$ or $\varepsilon$-clo$(q)$, is a set of states reachable from $q$ by $\varepsilon$-transitions.

$$E(q) = \varepsilon\text{-clo}(q) = \left\{ r \in Q \;\middle|\; q \xrightarrow{\varepsilon} r \right\}$$

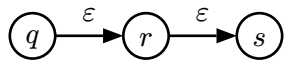
This definition can be extended to the *sets of states*. For $P \subseteq Q$:

$$E(P) = \bigcup_{q \in P} E(q)$$

# $\varepsilon$-NFA [2]

**Note**: $q \in \varepsilon\text{-clo}(q)$ by reflexivity: every state is reachable from itself by an $\varepsilon$-path of length 0.

*Example*: For the following NFA, epsilon closure of $q$ is $\varepsilon\text{-clo}(q) = \{q, r, s\}$.



# From $\varepsilon$-NFA to NFA

To construct an NFA from an $\varepsilon$-NFA:

1. Perform a *transitive closure* of $\varepsilon$-transitions.
   - After that, accepted words contain *no two consecutive* $\varepsilon$-transitions.
2. *Back-propagate* accepting states over $\varepsilon$-transitions.
   - After that, accepted words *do not end* with $\varepsilon$.
3. Perform *symbol-transition back-closure* over $\varepsilon$-transitions.
   - After that, accepted words *do not contain* $\varepsilon$-transitions.
4. *Remove* $\varepsilon$-transitions.
   - After that, you get a plain NFA.

# Rabin–Scott Powerset Construction

Any NFA can be converted to a DFA using *Rabin–Scott subset construction*.

$\mathcal{A}_{\mathrm{N}} = (Q_{\mathrm{N}}, \Sigma, \delta_{\mathrm{N}}, q_0, F_{\mathrm{N}})$

- $Q_{\mathrm{N}} = \{q_1, q_2, ..., q_n\}$
- $\delta_{\mathrm{N}} : Q_{\mathrm{N}} \times \Sigma \to \mathcal{P}(Q_{\mathrm{N}})$

$\mathcal{A}_{\mathrm{D}} = (Q_{\mathrm{D}}, \Sigma, \delta_{\mathrm{D}}, \{q_0\}, F_{\mathrm{D}})$

- $Q_{\mathrm{D}} = \mathcal{P}(Q_{\mathrm{N}}) = \{\emptyset, \{q_1\}, ..., \{q_2, q_4, q_5\}, ..., Q_{\mathrm{N}}\}$
- $\delta_{\mathrm{D}} : Q_{\mathrm{D}} \times \Sigma \to Q_{\mathrm{D}}$
- $\delta_{\mathrm{D}} : (A, c) \mapsto \{r \mid \exists q \in A.\, r \in \delta_{\mathrm{N}}(q, c)\}$
- $F_{\mathrm{D}} = \{A \mid A \cap F_{\mathrm{N}} \neq \emptyset\}$

**Warning:** The powerset construction can produce *exponentially many* states.

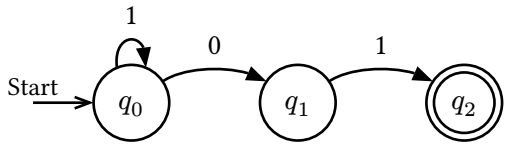
An NFA with $n$ states can require a DFA with up to $2^n$ states.

In practice, many of these states are unreachable and can be pruned.

# Converting an NFA to a DFA

We now examine a complete example of the powerset construction.

*Example*: Consider the NFA $\mathcal{N}$ over alphabet $\Sigma = \{0, 1\}$ that accepts strings ending with 01:




Formally, $\mathcal{N} = (Q, \Sigma, \delta, q_0, F)$ where:

- $Q = \{q_0, q_1, q_2\}$
- $\delta(q_0, 0) = \{q_0, q_1\}$, $\delta(q_0, 1) = \{q_0\}$, $\delta(q_1, 1) = \{q_2\}$, all other transitions go to $\emptyset$
- $q_0$ is start state
- $F = \{q_2\}$

## Reachable States

We begin with the start state of the DFA: $\{q_0\}$.

Compute transitions from $\{q_0\}$:

- On 0: $\delta_{D(\{q_0\},0)} = \delta_{N(q_0,0)} = \{q_0, q_1\}$
- On 1: $\delta_{D(\{q_0\},1)} = \delta_{N(q_0,1)} = \{q_0\}$

Now we have new states $\{q_0, q_1\}$ and $\{q_0\}$. $\{q_0\}$ is already known.

## New Transitions

From $\{q_0, q_1\}$:

- On 0: $\delta_{N(q_0,0)} \cup \delta_{N(q_1,0)} = \{q_0, q_1\} \cup \emptyset = \{q_0, q_1\}$
- On 1: $\delta_{N(q_0,1)} \cup \delta_{N(q_1,1)} = \{q_0\} \cup \{q_2\} = \{q_0, q_2\}$

New state: $\{q_0, q_2\}$

From $\{q_0, q_2\}$:

- On 0: $\delta_{N(q_0,0)} \cup \delta_{N(q_2,0)} = \{q_0, q_1\} \cup \emptyset = \{q_0, q_1\}$
- On 1: $\delta_{N(q_0,1)} \cup \delta_{N(q_2,1)} = \{q_0\} \cup \emptyset = \{q_0\}$

No new states.

## Accepting States

A DFA state is accepting if it contains any NFA accepting state:

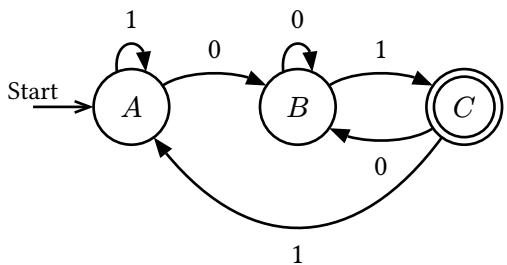
- $\{q_0\}$: contains $q_0$? No ($q_0 \notin F$). Not accepting.
- $\{q_0, q_1\}$: contains $q_2$? No. Not accepting.
- $\{q_0, q_2\}$: contains $q_2$? Yes ($q_2 \in F$). Accepting.

# The Resulting DFA

We have three reachable states: $A = \{q_0\}$, $B = \{q_0, q_1\}$, $C = \{q_0, q_2\}$.

Transition table:

| State | 0 | 1 | Accepting? |
|---|---|---|---|
| $A = \{q_0\}$ | $B$ | $A$ | ✗ |
| $B = \{q_0, q_1\}$ | $B$ | $C$ | ✗ |
| $C = \{q_0, q_2\}$ | $B$ | $A$ | ✓ |



# Verifying Equivalence

Both automata accept the same language: strings ending with 01.

- The NFA has 3 states, the DFA has 3 reachable states (out of possible $2^3 = 8$).
- Unreachable states like $\{q_1\}$, $\{q_2\}$, $\{q_1, q_2\}$, $\{q_0, q_1, q_2\}$, $\emptyset$ were never generated.

**Observation:** Although the powerset construction can produce exponentially many states, many of them are often unreachable. In this example, the resulting DFA is no larger than the original NFA.

So far, we have studied three different *representations* of languages: regular expressions, DFAs, and NFAs. A natural question arises: Are these representations equivalent? Can they express the same set of languages? Kleene's theorem answers this definitively.

# Kleene's Theorem

# Kleene's Theorem

**Theorem 2** (Kleene): $\mathrm{REG} = \mathrm{AUT}$.

That is, a language is *regular* (definable by a regular expression) if and only if it is recognized by a *finite automaton*.

**Proof** *(*$\mathrm{REG} \subseteq \mathrm{AUT}$*)*: *For every regular language, there is a DFA that recognizes it.*

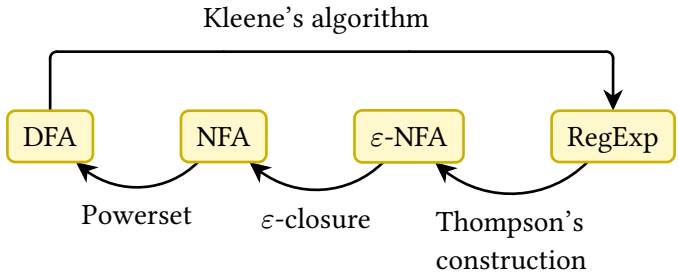
Use *Thompson's construction* to build an $\varepsilon$-NFA from a regular expression, and then convert it to a DFA. □

**Proof** *(*$\mathrm{AUT} \subseteq \mathrm{REG}$*)*: *The language recognized by a DFA is regular.*

Use *Kleene's algorithm* to construct a regular expression from an automaton. □

# The Equivalence Cycle

The diagram below shows the constructive conversions between the four equivalent representations. Each arrow denotes an explicit algorithm.




All four representations — DFA, NFA, $\varepsilon$-NFA, and regular expressions — are *equally powerful*.

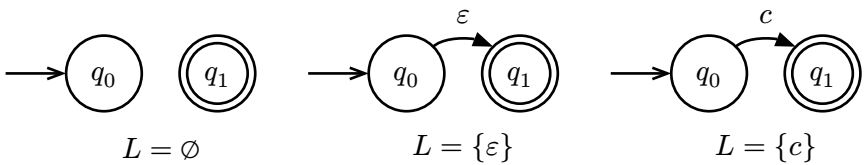
They describe exactly the same class of languages: the *regular languages*.

# Thompson's Construction

*Thompson's construction* is a method of constructing an $\varepsilon$-NFA from a regular expression.

Prove REG $\subseteq$ AUT by induction over the *generation index* $k$. Show that $\forall k.\, \mathrm{Reg}_k \subseteq \mathrm{AUT}$.

**Base:** $k = 0$, construct automata for $\mathrm{Reg}_0 = \{\emptyset, \{\varepsilon\}, \{c\} \text{ for } c \in \Sigma\}$.




**Induction step:** $k > 0$, already have automata $\mathcal{A}_1, \mathcal{A}_2$ for languages $L_1, L_2 \in \mathrm{Reg}_{k-1}$.

Construct automata for:

- $L_1 \cup L_2$ — add new start state with $\varepsilon$-transitions to both $\mathcal{A}_1$ and $\mathcal{A}_2$
- $L_1 \cdot L_2$ — connect accepting states of $\mathcal{A}_1$ via $\varepsilon$-transitions to start of $\mathcal{A}_2$
- $L_1^*$ — add new start/accept state with $\varepsilon$-transitions forming a loop through $\mathcal{A}_1$

# Kleene's Algorithm

**Definition 20**: *Kleene's algorithm* is a method of constructing a regular expression from a DFA.

Let $R_{ij}^k$ be a set of all words that take $\mathcal{A}$ from state $q_i$ to $q_j$ without going *through* (both entering and leaving) any state numbered higher than $k$. Note that $i$ and $j$ *can* be higher than $k$. Since all states are numbered 1 to $n$, $R_{ij}^n$ denotes the set of all words that take $q_i$ to $q_j$. We can define $R_{ij}^k$ recursively:

$$R_{ij}^k = R_{ij}^{k-1} \cup R_{ik}^{k-1} \left(R_{kk}^{k-1}\right)^* R_{kj}^{k-1}$$

$$R_{ij}^0 = \begin{cases} \{a \mid \delta(q_i, a) = q_j\} & \text{if } i \neq j \\ \{a \mid \delta(q_i, a) = q_j\} \cup \{\varepsilon\} & \text{if } i = j \end{cases}$$

Kleene's algorithm is structurally identical to the *Floyd–Warshall* algorithm for shortest paths in graphs. Both use dynamic programming with "allowed intermediate nodes up to $k$".

# Limits of Finite Automata

*"There are more things in heaven and earth, Horatio,*
*than are dreamt of in your philosophy."*

***— William Shakespeare***

# The Memory Bottleneck of Finite Automata

When does a language fail to be regular? Finite automata (DFAs and NFAs) possess exactly one form of memory: **their current state**.

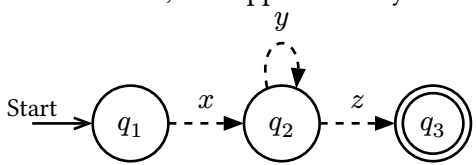
- A DFA with $n$ states can distinguish at most $n$ different "situations" at any point while reading a string.
- It **cannot count** indefinitely.
  (*e.g.*, verifying if there are exactly 1,000,000 zeros vs 1,000,001 zeros might require 1M states).
- It **cannot match** unbounded nested structures.
  (*e.g.*, balancing parentheses in code, checking `<html>` tags).

**Key insight:** If a language requires unbounded memory to recognize (for example, keeping track of an arbitrarily large number), it **cannot be regular**.

To formalize this, we exploit the simplest consequence of finite state spaces: the **Pigeonhole Principle**.

# Re-visiting States

- Let $D$ be a DFA with $n$ states.
- Any string $w$ accepted by $D$ that has length at least $n$ will force the DFA to take at least $n$ transitions.
- Taking $n$ transitions means visiting $n + 1$ states.
- By the *Pigeonhole Principle*, because the DFA only has $n$ distinct states, **at least one state must be visited twice**.
- Thus, the execution path contains a **loop**. The substring of $w$ that drives the DFA around this loop can be repeated (pumped) as many times as we want, or skipped entirely!




Informally:

- Let $L$ be a regular language.
- If we have a string $w \in L$ that is "sufficiently long", then we can *split* the string into *three pieces* and *"pump"* the middle.
- We can write $w = xyz$ such that $xy^0z$, $xy^1z$, $xy^2z$, ..., $xy^nz$, ... are all in $L$.

# Weak Pumping Lemma

**Theorem 3** (Weak Pumping Lemma for Regular Languages): Let $L$ be regular. Then there exists a purely structural constant $n \in \mathbb{N}$ (the **pumping length**), $n > 0$, such that for every "sufficiently long" $w \in L$ with $|w| \geq n$, there are strings $x, y, z$ where $w = xyz$, and:

1. $y \neq \varepsilon$ (the loop can't be empty)
2. For every $i \geq 0$, $xy^i z \in L$ (the loop can be skipped or repeated)

$$\underbrace{q_0 \cdots}_{x} \underbrace{q_k \cdots}_{y} \underbrace{q_k \cdots}_{z} q_f$$

Here, $x$ is the path to the loop, $y$ is the loop itself, $z$ is the path from the loop to the final state.

*Example*: Let $\Sigma = \{0, 1\}$ and $L = \{w \in \Sigma^* \mid w \text{ contains } 00 \text{ as a substring}\}$. Given any string of length $\geq 3$ in $L$, we can find a substring to pump. For instance, in $w = 100$, we let $x = 1, y = 0, z = 0$. The pumped strings $10^i 0$ still contain "00" safely!

*Example*: Let $\Sigma = \{0, 1\}$ and $L = \{\varepsilon, 0, 1, 00, 01, 10, 11\}$. The weak pumping lemma trivially holds with pumping length $n = 3$. There are **no** strings in $L$ of length $\geq 3$, so the condition "for every $w \in L$ with $|w| \geq 3$..." is vacuously true!

# The Equality Language

**Definition 21**: The *equality problem* is, given two strings $x$ and $y$, to decide whether $x = y$.

*Example*: Let $\Sigma = \{0, 1, \#\}$. We can *encode* the equality problem as a string of the form $x\#y$.

- "Is *001* equal to *110*?" would be $001\#110$.
- "Is *11* equal to *11*?" would be $11\#11$.

Let $\text{EQUAL} = \{w\#w \mid w \in \{0,1\}^*\}$.

**Question:** Is EQUAL a *regular* language?

**Strategy:** Assume regularity, choose $w = 0^n\#0^n$, then pump to destroy equality.

# EQUAL is Not Regular

**Theorem 4**: EQUAL is not a regular language.

**Proof**: By contradiction. Assume EQUAL is regular, and let $n$ be the pumping length.
Take $w = 0^n\#0^n \in \text{EQUAL}$. Write $w = xyz$ as in the weak pumping lemma.

The block $y$ cannot contain $\#$, because pumping with $i = 0$ would remove $\#$ and produce a string outside EQUAL. Hence $y$ lies entirely to the left or entirely to the right of $\#$, so $y = 0^k$ for some $k > 0$.

We now pump with $i = 2$:
Case 1: $y$ is to the left of $\#$. Then $xy^2z = 0^{n+k}\#0^n \notin \text{EQUAL}$.
Case 2: $y$ is to the right of $\#$. Then $xy^2z = 0^n\#0^{n+k} \notin \text{EQUAL}$.

In both cases we contradict the pumping lemma. Therefore EQUAL is not regular. □

A finite automaton cannot compare two unbounded strings for equality, because it has only finitely many states, *i.e.* it lacks "memory".

# The Classic Non-regular Language

*Example*: Consider the language $L = \{0^n 1^n \mid n \in \mathbb{N}\}$.

- $L = \{\varepsilon, 01, 0011, 000111, 00001111, ...\}$
- $L$ is a classic example of a non-regular language.
- **Intuitively:** if you have *only finitely many states* in a DFA, you cannot *"remember"* an arbitrary number of 0s to match *the same* number of 1s.

How would we prove that $L$ is non-regular?

Use the Pumping Lemma to show that $L$ *cannot* be regular.

# Pumping Lemma as a Game

The Pumping Lemma contains alternating quantifiers ($\exists, \forall, \exists, \forall$).

Any such statement can be framed naturally as a **two-player game**.

It is useful to view the lemma as a game between a **prover** and an **adversary**.

- The **prover** tries to force a contradiction and show that the language is not regular.
- The **adversary** tries to produce a legal pumping decomposition.

The interaction has four steps:

1. The adversary chooses a pumping length $n$.
2. The prover chooses a word $w \in L$ with $|w| \geq n$.
3. The adversary chooses a decomposition $w = xyz$ with $y \neq \varepsilon$.
4. The prover chooses $i \geq 0$ and tries to force $xy^iz \notin L$.

# Pumping Game for $L = \{0^n 1^n \mid n \in \mathbb{N}\}$

A complete play: the adversary and the prover alternate over the four moves.

1. **Adversary** picks the pumping length $n$.
2. **Prover** plays the witness $w = 0^n 1^n \in L$, of length $|w| = 2n \geq n$.
3. **Adversary** chooses a legal split $w = xyz$ with $y \neq \varepsilon$.
4. **Prover** replies with a pumping index $i$ that defeats the split: $i = 0$ when $y$ consists only of 0s or only of 1s (then $xz$ has unequal counts), and $i = 2$ when $y$ straddles the boundary between 0s and 1s (then $xy^2z$ is no longer of the form $0^*1^*$). In either case $xy^iz \notin L$.

**Verdict — the prover wins.**

For any split the adversary can choose, the prover holds a reply $i$ that pumps the word out of $L$.

The pumping conclusion therefore fails, which proves that $L = \{0^n 1^n \mid n \in \mathbb{N}\}$ is *not regular*.

# Proof that $0^n1^n$ is Not Regular

**Theorem 5**: $L = \{0^n1^n \mid n \in \mathbb{N}\}$ is not regular.

**Proof**: By contradiction. Assume that $L$ is regular.

Let $n$ be the pumping length guaranteed by the weak pumping lemma ("there exists $n$..."). Consider the string $w = 0^n1^n$. Then $|w| = 2n \geq n$ and $w \in L$, so we can write (split) $w = xyz$ such that $y \neq \varepsilon$ and for any $i \in \mathbb{N}$, we have $xy^iz \in L$.

Case 1: $y$ consists solely of 0s. Then $xy^0z = xz = 0^{n-|y|}1^n$.
Since $|y| > 0$, the number of 0s is less than $n$, so $xz \notin L$.

Case 2: $y$ consists solely of 1s. Then $xy^0z = xz = 0^n1^{n-|y|}$.
Since $|y| > 0$, the number of 1s is less than $n$, so $xz \notin L$.

Case 3: $y$ consists of some 0s followed by some 1s. If we pump $y$ with $i = 2$, then $y^2 = yy$, which places 1s from the first copy of $y$ **before** the 0s of the second copy.
Thus the resulting string $xy^2z$ is not even of the form $0^*1^*$, so it cannot be in $L$.

## Proof that $0^n1^n$ is Not Regular [2]

In all three possible ways the adversary splits the string, we found a choice of $i$ giving a string outside $L$. This contradicts the weak pumping lemma. Thus $L$ is **not regular**. □

# Cautionary Note

- The pumping lemma describes a *necessary* condition for regularity.
  - Regularity $\rightarrow$ Pumping Lemma holds.
  - Pumping Lemma FAILS $\rightarrow$ Not Regular (by contrapositive).
- It is *not a sufficient* condition.
  - Pumping Lemma holds $\not\rightarrow$ Regularity.
  - There exist non-regular languages that still satisfy the pumping lemma.

**Warning:** "The language satisfies the pumping lemma, therefore it is regular."

This is a logic error. The pumping lemma is a tool for disproving regularity, not for establishing it. To prove that a language is regular, one must construct a DFA, NFA, or regular expression.

# The Full Pumping Lemma

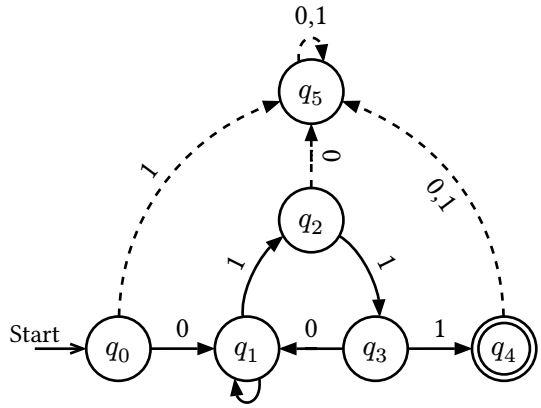
By the pigeonhole principle: any accepting path visiting $n+1$ states must revisit some state — and this state repeat occurs *within the first $n$ steps*.

**Theorem 6** (Full Pumping Lemma for Regular Languages): Let $L$ be a regular language. Then there exists $n \in \mathbb{N}$, $n > 0$, such that for every $w \in L$ with $|w| \geq n$, there exist strings $x, y, z$ with:

- $w = xyz$,
- $y \neq \varepsilon$,
- $|xy| \leq n$, and
- for every $i \in \mathbb{N}$, $xy^i z \in L$.

**Upgrade from the weak version:** the additional constraint $|xy| \leq n$ forces the pump $y$ to lie within the first $n$ characters of $w$. This closes a loophole: some non-regular languages (such as $\{w \mid w \text{ has equal 0s and 1s}\}$) pass the *weak* lemma, but are caught by the *full* version.

# A DFA View of the Pumping Lemma




$$q_0 \underbrace{\xrightarrow{0}}_{x=0} q_1 \underbrace{\xrightarrow{1} q_2 \xrightarrow{1} q_3 \xrightarrow{0}}_{y=110} q_1 \underbrace{\xrightarrow{1} q_2 \xrightarrow{1} q_3 \xrightarrow{0}}_{y=110} q_1 \underbrace{\xrightarrow{1} q_2 \xrightarrow{1} q_3 \xrightarrow{0}}_{y=110} q_1 \underbrace{\xrightarrow{1} q_2 \xrightarrow{1} q_3 \xrightarrow{1}}_{z=111} q_4$$

**Note**: The first repeated state is $q_1$, revisited on step 4, so $y = 110$ lies within the first $n = 6$ symbols. Here $x = 0$, and pumping $y$ preserves acceptance until the suffix $z$ reaches $q_4$.

# Equal Numbers of 0s and 1s

Consider the language $L$ over $\Sigma = \{0, 1\}$ of strings $w \in \Sigma^*$ that contain *an equal number* of 0s and 1s.

For example, `01` in $L$, `11011` not in $L$, `110010` in $L$.

**Theorem 7**: $L = \{w \in \{0,1\}^* \mid w \text{ has an equal number of 0s and 1s}\}$ is *not regular*.

**Proof**: By contradiction. Assume that $L$ is regular.

Let $n$ be the pumping length guaranteed by the full pumping lemma. Consider the string $w = 0^n 1^n$. Then $|w| = 2n \geq n$ and $w \in L$. Therefore, there exist strings $x$, $y$, and $z$ such that $w = xyz$, $|xy| \leq n$, $y \neq \varepsilon$, and for any $i \in \mathbb{N}$, we have $xy^i z \in L$.

Since $|xy| \leq n$, $y$ must consist solely of 0s. But then $xy^2z = 0^{n+|y|}1^n$, and since $|y| > 0$, $xy^2z \notin L$.

We have reached a contradiction, so our assumption was wrong and $L$ is not regular. □

**Note**: This language *cannot* be shown to be non-regular using the *weak* pumping lemma alone (it passes the weak version!). The constraint $|xy| \leq n$ from the full version is essential.

# Palindromes Over $\{0, 1\}$

**Definition 22**: A *palindrome* is a string that reads the same forwards and backwards.

Let $\mathrm{PAL} = \{w \in \{0,1\}^* \mid w = w^R\}$ where $w^R$ is the reversal of $w$.

**Theorem 8**: PAL is not regular.

**Proof**: Assume PAL is regular, with pumping length $n$. Choose the string $w = 0^n 10^n$. Then $w \in \mathrm{PAL}$ and $|w| = 2n + 1 \geq n$.

By the full pumping lemma, there exist strings $x, y, z$ such that $w = xyz$ and:

- $y \neq \varepsilon$ (the loop is real)
- $|xy| \leq n$ (the loop happens entirely within the first $n$ characters)

Since the first $n$ characters of $w$ are all 0s, the string $y$ must be $0^k$ for some $k > 0$. Choosing $i = 2$ gives $xy^2z = 0^{n+k}10^n$, which is not a palindrome. This contradicts the lemma, so PAL is not regular. □

# Palindromes Over $\{0, 1\}$ [2]

*Example*: For $n = 3$: the string 000100 is a palindrome. The pumping lemma gives us $xyz$ with $|xy| \leq 3$. The $y$ part is some prefix of 000, say $y = 00$. Then:

- $xyz$ (original): 000100 ✓ (palindrome)
- $xy^2z$ (pumped): 00000100 ✗ (not a palindrome: 5 leading zeros but only 2 trailing zeros)

# Strings with More 0s than 1s

**Definition 23**: Let $\text{MORE0} = \{w \in \{0,1\}^* \mid w \text{ has strictly more 0s than 1s}\}$.

**Theorem 9**: MORE0 is not regular.

**Proof**: Assume regularity, and let $n$ be the pumping length. Choose the string $w = 0^{n+1}1^n$.

- It clearly has $n+1 > n$ zeros, so $w \in \text{MORE0}$.
- Its length $2n+1 \geq n$, so it works.

The adversary splits $w = xyz$ such that $|xy| \leq n$. The first $n$ characters of $w$ are all 0s. Thus, $y$ captures exactly $k$ zeros (where $k > 0$).

If we choose $i = 2$, the number of 0s increases, so that does not help. Instead, choose $i = 0$ and delete the block $y$. Our modified string is $xz = 0^{n+1-k}1^n$. Since $k \geq 1$, the remaining zeros are at most $n$.

The new string has at most $n$ zeros, but it still has exactly $n$ ones. Therefore it does not have strictly more 0 s than 1s, so it is not in MORE0. This is a contradiction. □

# Strings with Unequal Counts

**Definition 24**: Let $\text{UNEQUAL} = \{w \in \{0,1\}^* \mid w \text{ has different number of 0s and 1s}\}$.

**Theorem 10**: UNEQUAL is not regular.

**Proof**: Note that $\text{UNEQUAL} = \overline{\text{EQUAL}}$, the complement of the language of strings with equal 0s and 1s. Since EQUAL is not regular (proved earlier), and regular languages are closed under complement, UNEQUAL cannot be regular either. (If UNEQUAL were regular, its complement EQUAL would also be regular — contradiction.) □

# A Complex-Looking Regular Language

**Definition 25**: Let $\text{DIV7} = \{w \in \{0,1\}^* \mid w \text{ interpreted as binary number is divisible by } 7\}$.

**Theorem 11**: DIV7 is regular.

**Proof**: Construct a DFA with 7 states representing the remainder modulo 7. As we read each bit $b \in \{0,1\}$, update the remainder: $r_{\text{new}} = (2 \cdot r_{\text{old}} + b) \bmod 7$. Accept if final remainder is 0. □

**Key insight:** The pumping lemma proves non-regularity. Some languages that seem complex are actually regular because they require only *bounded memory*. Here, the 7 states encode the remainder — that's enough. EQUAL requires *unbounded* memory (count of 0s), so it cannot be regular.

# Intuition Behind the Myhill-Nerode Theorem

The pumping lemma studies *loops* in long computations. The Myhill-Nerode theorem studies something even more basic: when two prefixes are *indistinguishable* to a finite automaton.

Suppose a DFA has read two different prefixes $x$ and $y$ and, after each of them, ends in the same state. From that moment on, the machine has exactly the same future behavior on both inputs. No suffix can reveal which prefix was read earlier, because the automaton remembers only its current state.

**Key insight:** If a language forces the machine to keep infinitely many different prefixes apart, then no finite automaton can recognize it.

# The Equivalence Relation

**Definition 26**: Let $L \subseteq \Sigma^*$. Define a relation $\equiv_L$ on $\Sigma^*$ by

$$x \equiv_L y \quad \text{iff} \quad \forall z \in \Sigma^*. (xz \in L \iff yz \in L).$$

Thus $x \equiv_L y$ means that *every* suffix $z$ has the same effect on $x$ and on $y$. In other words, $x$ and $y$ are indistinguishable with respect to membership in $L$.

*Example*: Let $L = \{w \in \{0,1\}^* \mid w \text{ ends with } 01\}$.

- The strings 001 and 101 are equivalent: both already end with 01, so after appending any suffix $z$, the membership of $001z$ and $101z$ depends only on $z$, not on the earlier prefix.
- The strings 0 and 01 are distinguishable: take $z = \varepsilon$. Then $0z = 0 \notin L$, but $01z = 01 \in L$.

# The Myhill-Nerode Theorem

**Theorem 12** (Myhill-Nerode Theorem): A language $L$ is regular if and only if the relation $\equiv_L$ has finite index, that is, only finitely many equivalence classes.

Moreover, the number of equivalence classes is exactly the number of states in the minimal DFA for $L$.

**Proof** *(idea)*: *($\Rightarrow$)* Assume $L$ is regular, and let a DFA $A$ with state set $Q$ recognize it. Every string $w \in \Sigma^*$ sends the start state to one state of $Q$. If two strings $x$ and $y$ lead to the same state, then for every suffix $z$ the automaton behaves identically on $xz$ and $yz$. Hence $x \equiv_L y$. Therefore there can be at most $|Q|$ equivalence classes.

*($\Leftarrow$)* Now assume $\equiv_L$ has only finitely many equivalence classes. Build a DFA whose states are those classes. The start state is the class of $\varepsilon$. On symbol $a$, move from the class of $x$ to the class of $xa$. Accept exactly those classes containing strings from $L$. This automaton recognizes $L$, so $L$ is regular. □

# Connection to Minimal DFAs

The theorem does more than characterize regular languages: it explains what the states of the minimal DFA *mean*. Each state corresponds to one equivalence class of prefixes.

**Definition 27**: To construct the minimal DFA for $L$:

1. Compute the equivalence classes of $\equiv_L$.
2. Use these classes as the states.
3. Take the class of $\varepsilon$ as the start state.
4. Mark a class as accepting iff it contains a string from $L$.
5. On input $a$, send the class of $x$ to the class of $xa$.

*Example*: Let $L = \{w \mid w \text{ has an odd number of 1s}\}$ over $\{0, 1\}$. There are exactly two equivalence classes:

- prefixes with an even number of 1s;
- prefixes with an odd number of 1s.

Therefore the minimal DFA has exactly two states.

# Proof via Infinite Distinguishability

To prove that a language is *not* regular using Myhill-Nerode, it is enough to exhibit infinitely many pairwise distinguishable strings.

**Definition 28**: A family of strings $x_0, x_1, x_2, \ldots$ is *pairwise $L$-distinguishable* if for every $i \neq j$ there exists a suffix $z$ such that exactly one of $x_i z$ and $x_j z$ belongs to $L$.

**Theorem 13**: If $L$ contains an infinite pairwise $L$-distinguishable family, then $\equiv_L$ has infinite index. Consequently, $L$ is not regular.

**Proof**: If the strings are pairwise distinguishable, then no two of them can belong to the same equivalence class of $\equiv_L$. Hence infinitely many different strings yield infinitely many different equivalence classes. By the Myhill-Nerode theorem, a regular language can have only finitely many such classes. Therefore $L$ is not regular. □

# Proof via Infinite Distinguishability [2]

*Example ($\{0^n 1^n \mid n \geq 0\}$ is not regular)*: Consider the infinite family $x_i = 0^i$ for $i \in \mathbb{N}$.

Take two distinct indices $i < j$. Choose the suffix $z = 1^i$. Then

- $x_i z = 0^i 1^i \in L$,
- $x_j z = 0^j 1^i \notin L$.

Thus $x_i$ and $x_j$ are distinguishable. Since this works for every pair $i \neq j$, the family is infinite and pairwise distinguishable. Therefore $L$ is not regular.

*Example ($\{ww \mid w \in \{0,1\}^*\}$ is not regular)*: Consider the family $x_i = 0^i 1$ for $i \in \mathbb{N}$.

Take two indices $i < j$ and choose the suffix $z = 0^i 1$. Then

- $x_i z = 0^i 10^i 1 = (0^i 1)(0^i 1) \in L$,
- $x_j z = 0^j 10^i 1 \notin L$.

Hence the strings $x_i$ are pairwise distinguishable, so the language is not regular.

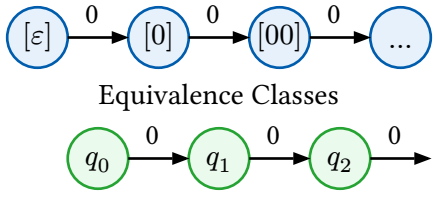
# Pumping Lemma and Myhill-Nerode

| Method | Strength | Application |
|---|---|---|
| Pumping Lemma | Necessary condition only | Can disprove regularity, cannot prove it |
| Myhill-Nerode | Necessary and sufficient | Can both prove and disprove regularity |
| Pumping Lemma | Constructive counterexample | Adversary splits, you choose pump value |
| Myhill-Nerode | Structural characterization | Exhibit infinite distinguishable set |
| Pumping Lemma | Based on pigeonhole principle | Focuses on long strings in the language |
| Myhill-Nerode | Based on equivalence relations | Focuses on distinguishing extensions |

**Why this matters:** While the pumping lemma is taught first due to its simpler structure, Myhill-Nerode provides deeper insight into *why* some languages aren't regular and connects directly to the minimal automaton structure.

# Visualizing Equivalence Classes

For a regular language, the equivalence classes correspond to states in the minimal DFA:




| Diagram Part | Meaning |
|---|---|
| Top row: $[\varepsilon], [0], [00], \ldots$ | Myhill-Nerode equivalence classes of prefixes |
| Bottom row: $q_0, q_1, q_2$ | States of the minimal DFA |
| Arrows labeled 0 | Appending symbol 0 and moving to the class of the extended prefix |

**Note**: The same principle works for every symbol in the alphabet; only symbol 0 is drawn to keep the picture readable.

# Closure and Decision Properties

# Closure of Regular Languages

**Theorem 14**: The class REG is closed under all the following operations:

1. The *union* of two regular languages is regular.
2. The *intersection* of two regular languages is regular.
3. The *complement* of a regular language is regular.
4. The *difference* of two regular languages is regular.
5. The *reversal* of a regular language is regular.
6. The *Kleene star* of a regular language is regular.
7. The *concatenation* of regular languages is regular.
8. A *homomorphism* (substitution of strings for symbols) of a regular language is regular.
9. The *inverse homomorphism* of a regular language is regular.

Closure properties are a *powerful tool* for proving languages are (or are not) regular. Instead of building automata from scratch, we can *compose* known regular languages using these operations.

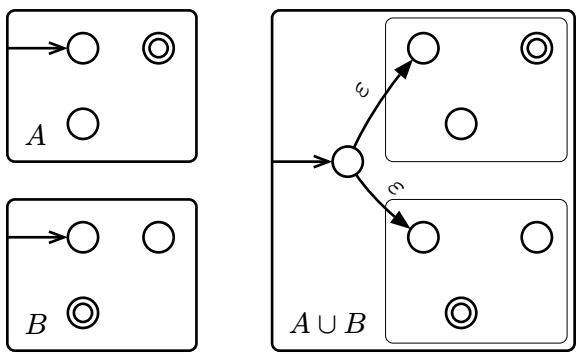
# Closure under Union

**Theorem 15**: If $L$ and $M$ are regular languages, then so is their union $L \cup M$.

**Proof**: Since $L$ and $M$ are regular, they have regular expressions, *i.e.* $L = \mathcal{L}(R)$ and $M = \mathcal{L}(S)$.

Then $L \cup M = \mathcal{L}(R + S)$ by the definition of the union $(+)$ operator for regular expressions. □

**Note**: This proof is "algebraic" (via regular expressions). The picture below gives the automata-construction intuition.

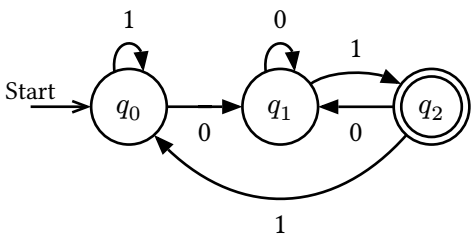


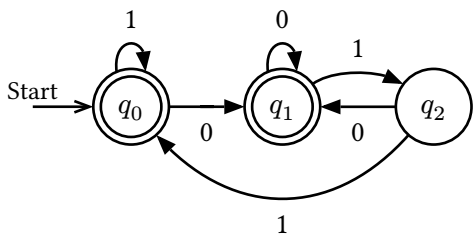
# Closure under Complement

**Theorem 16**: If $L$ is a regular language over the alphabet $\Sigma$, then its complement $\overline{L} = \Sigma^* - L$ is also a regular language.

**Proof**: Let $L = \mathcal{L}(A)$ for some DFA $A = (Q, \Sigma, \delta, q_0, F)$. Then $\overline{L} = \mathcal{L}(B)$, where $B$ is the DFA $(Q, \Sigma, \delta, q_0, Q - F)$. That is, $B$ is exactly like $A$, but with the accepting states *flipped*. Then $w$ is in $\overline{L}$ if and only if $\delta(q_0, w)$ is in $Q - F$, which occurs if and only if $w$ is not in $\mathcal{L}(A)$. □

*Example*: The DFA $A$ below on the left accepts only the strings of 0's and 1's that end in `01`, $\mathcal{L}(A) =$ `(0+1)*01`. The complement of $\mathcal{L}(A)$ is all strings of 0's and 1's that *do not* end in `01`. Below on the right is the automaton for $\{0,1\}^* - \mathcal{L}(A)$.



# Closure under Intersection

**Theorem 17**: If $L$ and $M$ are regular languages, then so is their intersection $L \cap M$.

**Proof** *(simple)*: $L \cap M = \overline{\overline{L} \cup \overline{M}}$. □

**Proof**: We can directly construct a "product" DFA for the intersection of two regular languages.

Let $L$ and $M$ be the languages of automata $A_L = (Q_L, \Sigma, \delta_L, q_L, F_L)$ and $A_M = (Q_M, \Sigma, \delta_M, q_M, F_M)$.

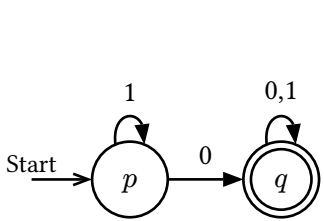
For $L \cap M$, we construct the automaton $A$ that simulates both $A_L$ and $A_M$. The states of $A$ are the product of the states of $A_L$ and $A_M$. The initial state is $(q_L, q_M)$, and the accepting states are $F_L \times F_M$. The transitions: $\delta(\langle p, q \rangle, c) = \langle \delta_L(p, c), \delta_M(q, c) \rangle$.

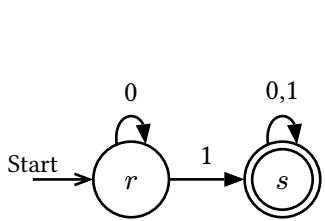
$A$ accepts $w$ iff both $A_L$ and $A_M$ accept $w$. □

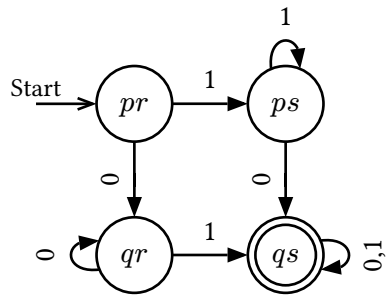
**Note**: De Morgan's identity gives a short proof; the product construction gives an explicit algorithm.

# Product Construction for Intersection

*Example*: The first automaton accepts all strings that *have a 0*. The second accepts all strings that *have a 1*. The product automaton accepts the *intersection*: all strings that *have both a 0 and a 1*.



# Closure under Difference and Reversal

**Theorem 18**: If $L$ and $M$ are regular languages, then so is their difference $L - M$.

**Proof**: $L - M = L \cap \overline{M}$. By previous theorems, $\overline{M}$ is regular, and $L \cap \overline{M}$ is also regular. □

**Definition 29**: The *reversal* of a string $w = a_1 a_2 ... a_n$ is the string $w^R = a_n a_{n-1} ... a_1$.

**Definition 30**: The *reversal* of a language $L$ is the language $L^R = \{w^R \mid w \in L\}$.

**Theorem 19**: If $L$ is a regular language, then so is its reversal $L^R$.

**Proof**: Structural induction on the regular expression $E$ defining $L$:

- *Basis:* If $E$ is $\varepsilon$, $\emptyset$, or $\mathtt{a}$, then $E^R = E$.
- *Induction:* $E = E_1 + E_2 \Rightarrow E^R = E_1^R + E_2^R$; $E = E_1 E_2 \Rightarrow E^R = E_2^R E_1^R$; $E = E_1^* \Rightarrow E^R = (E_1^R)^*$. □

# Decision Properties

**Converting among representations**

- $\varepsilon$-closure: $\mathcal{O}(n^3)$
- $\varepsilon$-NFA to DFA: $\mathcal{O}(n^3 2^n)$
- DFA to $\varepsilon$-NFA: $\mathcal{O}(n)$
- $\varepsilon$-NFA to RegEx: $\mathcal{O}(n^3 4^n)$
- RegEx to $\varepsilon$-NFA: $\mathcal{O}(n)$

Representation changes are computable but can be exponentially expensive in the worst case.

# Decidable Questions

**Decidable questions about REG**

1. Is the language *empty*? $\mathcal{O}(n^2)$
2. Is the language *finite*? $\mathcal{O}(n^2)$
3. Is $w$ *in* the language? $\mathcal{O}(|w|)$ for DFAs
4. Is $L \subseteq M$? Decidable
5. Is $L = M$? Decidable

# Threshold Theorems

**Theorem 20**: The language $L$ accepted by a finite automaton with $n$ states is *non-empty* iff the finite automaton accepts a word of length less than $n$.

**Theorem 21**: The language $L$ accepted by a finite automaton $M$ with $n$ states is *infinite* iff the automaton accepts some word of length $l$, where $n \leq l < 2n$.

# Summary of Regular Languages

1. **Equivalence** (Kleene's Theorem): DFA = NFA = $\varepsilon$-NFA = Regular Expression. Different representations, same class.
2. **Pumping Lemma:** if $L$ is regular with DFA of $n$ states, every $w \in L$ with $|w| \geq n$ can be pumped: $w = xyz$, $|y| \geq 1$, $|xy| \leq n$, $xy^iz \in L$ for all $i \geq 0$. A *necessary* condition only.
3. **Myhill–Nerode Theorem:** $L$ is regular iff the equivalence $x \equiv_L y$ ("$x$ and $y$ lead to the same future") has *finitely many classes*. This is both necessary and sufficient, and characterizes the minimal DFA.
4. **Closure:** regular languages are closed under union, intersection, complement, concatenation, Kleene star, reversal, homomorphism, and inverse homomorphism.
5. **All key questions are decidable:** emptiness, finiteness, membership, subset, equality.

**Hard ceiling:** a DFA has only *constant memory* (finitely many states). It cannot count, match, or track unbounded structure. Any language requiring counting or matching at arbitrary depth is *not* regular. Canonical witnesses: $\{a^nb^n\}$, $\{ww\}$, balanced parentheses.

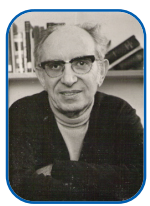
# Beyond Regular Languages

*"A grammar is a device that specifies the infinite set of well-formed sentences of a language and gives a structural description of each."*

***— Noam Chomsky***


John Backus


Peter Naur


Yehoshua Bar-Hillel


Aad van Wijngaarden


Andrey Terekhov


Niklaus Wirth


John McCarthy

# What Regular Languages Cannot Do

We have seen several non-regular languages:

- $L = \{0^n 1^n \mid n \in \mathbb{N}\}$ — requires *counting* to match 0s and 1s.
- $\text{EQUAL} = \{w\#w \mid w \in \{0,1\}^*\}$ — requires *remembering* an entire string.
- $L = \{w \in \{0,1\}^* \mid w \text{ has equal number of 0s and 1s}\}$ — requires a *counter*.

The common pattern: regular languages cannot *count* beyond a bounded amount. A DFA has only *finitely many states*, so it cannot track *unbounded* quantities.

**Fundamental limitation:** A finite automaton is a machine with *constant memory*. It can remember only a *bounded* amount of information about the input it has read.

**Upgrade:** give the machine a *stack* (last-in first-out memory). This gives the next level of computational power in the Chomsky hierarchy: *context-free languages*.

# Context-Free Languages

Regular languages cannot describe $\{a^n b^n \mid n \geq 0\}$: any DFA only has finite memory.

The key idea is to allow *recursive* rules — a grammar that can expand itself to arbitrary depth.

**Definition 31**: A *context-free grammar* (CFG) is a 4-tuple $G = (V, \Sigma, R, S)$ where:

- $V$ is a finite set of *variables* (non-terminals),
- $\Sigma$ is a finite set of *terminals* (disjoint from $V$),
- $R \subseteq V \times (V \cup \Sigma)^*$ is a finite set of *production rules* $A \to \alpha$,
- $S \in V$ is the *start symbol*.

The language of $G$ is the set of all terminal strings derivable from $S$:

$$\mathcal{L}(G) = \{w \in \Sigma^* \mid S \Rightarrow^* w\}$$

*Example*: Grammar $G$: $S \to 0S1 \mid \varepsilon$ generates $L = \{0^n 1^n \mid n \geq 0\}$.

$S \Rightarrow 0S1 \Rightarrow 00S11 \Rightarrow 000S111 \Rightarrow 000\varepsilon 111 = 000111$

# BNF and EBNF

*Backus–Naur Form* (BNF) is the standard notation for writing CFGs in language specifications — it is exactly the same formalism as CFGs, just a different syntax.

**Definition 32**: A *BNF rule* has the form:

$$\langle \text{symbol} \rangle \quad ::= \quad \alpha_1 \quad | \quad \alpha_2 \quad | \quad \ldots \quad | \quad \alpha_k$$

- $\langle \text{symbol} \rangle$ is a *non-terminal* (written in angle brackets); corresponds to a CFG variable.
- Each $\alpha_i$ is a sequence of *terminals* (quoted strings) and non-terminals.
- "::=" means "is defined as".
- "|" separates alternatives.

*Example*: BNF grammar for arithmetic expressions (from the ALGOL 60 report, Backus and Naur, 1960):

```
<expr>   ::= <expr> "+" <term>  |  <term>
<term>   ::= <term> "*" <factor>  |  <factor>
<factor> ::= "(" <expr> ")"  |  <id>
```

# BNF and EBNF [2]

The stratified structure encodes *precedence*: * binds tighter than +. This is exactly the unambiguous grammar $E \to E + T \mid T, T \to T \times F \mid F, F \to (E) \mid \texttt{id}$.

*Extended BNF* (EBNF) adds shorthand notation — no new expressive power, but much more readable:

| EBNF syntax | Meaning | BNF equivalent |
|---|---|---|
| $[\alpha]$ | Optional: zero or one $\alpha$ | $S \to \alpha \mid \varepsilon$ |
| $\{\alpha\}$ or $\alpha^*$ | Repetition: zero or more $\alpha$ | $S \to \alpha S \mid \varepsilon$ |
| $\alpha^+$ | One or more $\alpha$ | $S \to \alpha\alpha^*$ |
| $(\alpha_1 \mid \alpha_2)$ | Grouping with alternatives | Inline alternative |

**Where you'll encounter BNF/EBNF:** language reference manuals (Python, Rust, C), RFC protocol specifications, ISO SQL standard, JSON schema. Every modern compiler processes a grammar defined in BNF or EBNF.

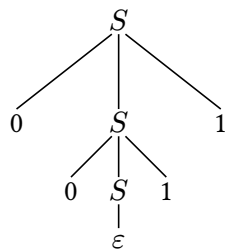
# Derivations and Parse Trees

**Definition 33**: A *derivation step* $\alpha A \beta \Rightarrow \alpha \gamma \beta$ applies production $A \to \gamma \in R$, replacing one variable $A$. *Derives* $\Rightarrow^*$ is the reflexive-transitive closure: zero or more steps. A *leftmost derivation* always replaces the leftmost variable first.

The same derivation information can be rendered as a *parse tree*:

**Derivation** of 0011 from $S \to 0S1 \mid \varepsilon$:

$$\begin{aligned} S &\Rightarrow 0S1 \\ &\Rightarrow 00S11 \\ &\Rightarrow 00\varepsilon 11 = 0011 \end{aligned}$$




Each step expands one node. The *leaves* (left to right) spell out the derived word.

**Note**: Compilers use parse trees as the *abstract syntax tree* (AST) of source code. Different derivation *orders* (leftmost, rightmost) produce identical parse trees in unambiguous grammars.

# Further CFG Examples

| Productions | Language | Application |
|---|---|---|
| $S \to 0S1 \mid \varepsilon$ | $\{0^n 1^n \mid n \geq 0\}$ | Classic matched pairs |
| $S \to aSa \mid bSb \mid a \mid b \mid \varepsilon$ | Palindromes over $\{a, b\}^*$ | Hashing, bioinformatics |
| $S \to SS \mid (\ S\ ) \mid \varepsilon$ | Balanced parentheses $\{(), (()), ()(), \ldots\}$ | Lisp, expression parsers |
| $E \to E + T \mid T$<br>$T \to T \times F \mid F$<br>$F \to (E) \mid \texttt{id}$ | Arithmetic expressions with correct precedence | Every compiler (real PLs) |
| $S \to aSb \mid bSa \mid SS \mid \varepsilon$ | $\{w \in \{a, b\}^* \mid \#_a(w) = \#_b(w)\}$ | Balance counting |

**Pattern:** A grammar variable acts as a "placeholder" carrying context through recursive structure — exactly what a *stack* does at runtime. This deep connection is Theorem: CFG = PDA.

# Ambiguous Grammars

**Definition 34**: A grammar $G$ is *ambiguous* if some word $w \in \mathcal{L}(G)$ has two distinct *leftmost derivations* (equivalently, two distinct parse trees).

*Example*: Grammar $E \to E + E \mid E \times E \mid (E) \mid \texttt{id}$ is *ambiguous*. The word $\texttt{id} + \texttt{id} \times \texttt{id}$ has two parse trees with **different** root operators:

- **Tree A** ($\times$ at root): interprets as $(\texttt{id} + \texttt{id}) \times \texttt{id}$ — wrong precedence!
- **Tree B** ($+$ at root): interprets as $\texttt{id} + (\texttt{id} \times \texttt{id})$ — correct!

Both are valid derivations in $E \to E + E \mid E \times E \mid \ldots$, so the grammar cannot determine which.

The fix: *stratify* the grammar so that higher-precedence operators are at deeper nesting levels.

$$E \to E + T \mid T, \quad T \to T \times F \mid F, \quad F \to (E) \mid \texttt{id}$$

Here $T$ (term) binds tighter than $E$ (expression): $\times$ before $+$. The stratified grammar has exactly *one* parse tree per expression, reflecting the intended meaning.

# Consequences of Ambiguous Grammars

**Inherently ambiguous languages**: A *context-free language* $L$ is *inherently ambiguous* if **every** CFG for $L$ is ambiguous. This means no clever reformulation of the grammar can remove the ambiguity.

Classic example: $L = \{a^i b^j c^k \mid i = j\} \cup \{a^i b^j c^k \mid j = k\}$. Any CFG for this union must handle both constraints simultaneously, leading to unavoidable ambiguity on words like $a^n b^n c^n$ (which are in the language for *both* reasons).

**Compiler engineering:** Parser generators (yacc, Bison, ANTLR) *reject* or *warn* on ambiguous grammars. Ambiguity manifests as shift/reduce or reduce/reduce *conflicts* in LR parsing tables. Resolving them requires either grammar rewriting or explicit precedence/associativity declarations.

**Decidability:** "Is grammar $G$ ambiguous?" is *undecidable* — there is no algorithm that can check all grammars. Modern tools use heuristics and partial analyses.

# Chomsky Normal Form

**Definition 35**: A CFG $G$ is in *Chomsky Normal Form* (CNF) if every production rule has one of two forms:

- $A \to BC$ where $B, C \in V$ (two variables), or
- $A \to a$ where $a \in \Sigma$ (a single terminal).

The only $\varepsilon$-production allowed is $S \to \varepsilon$ (only if $\varepsilon \in \mathcal{L}(G)$), and $S$ must not appear on any rule's right-hand side.

**Theorem 22**: Every context-free language has a CNF grammar.

CNF is a canonical form: it looks restrictive (no rule has more than two symbols on the right), but any CFL can be expressed in it without changing the language.

# Chomsky Normal Form [2]

**Why CNF exists:** Many algorithms and proofs about CFLs become significantly simpler when the grammar has a fixed, uniform shape.

Two key applications:

1. **CYK membership algorithm:** $\mathcal{O}(n^3)$ dynamic programming, only works on CNF grammars.
2. **Pumping Lemma proof:** The repeated variable on a long parse tree path is found via binary tree height bounds.

# CNF Properties

In a CNF parse tree, every internal node is either a *binary branching* ($A \to BC$) or a *terminal* ($A \to a$). This gives a clean structural guarantee:

**Theorem 23**: In a CNF parse tree for a word of length $n$, there are *exactly* $2n - 1$ internal nodes.

This means: the tree height is at most $2n - 1$, and a tree with height $\geq |V| + 1$ must contain a repeated variable on some root-to-leaf path (by Pigeonhole) — the key to the Pumping Lemma proof.

# CYK Algorithm

**CYK algorithm** (Cocke–Younger–Kasami): given CNF grammar $G$ and string $w = a_1 \ldots a_n$, fill a triangular table $T[i][j]$ = set of variables that derive $a_i a_{i+1} \ldots a_j$.

- Base: $T[i][i] = \{A \mid A \to a_i \in R\}$
- Step: $T[i][j] = \{A \mid A \to BC \in R, B \in T[i][k], C \in T[k+1][j] \text{ for some } k\}$
- Accept iff $S \in T[1][n]$.

Running time: $\mathcal{O}(n^3 \cdot |G|)$.

**Historical note:** CYK was the first polynomial-time CFL membership algorithm, developed independently by Cocke (1960), Younger (1967), and Kasami (1966). It is the CFL analogue of Floyd–Warshall for shortest paths.

# CNF Conversion Algorithm

Any CFG can be mechanically transformed into CNF in five steps (applied in order):

1. **New start symbol.** Add $S_0 \to S$ and make $S_0$ the start. Ensures $S$ never appears on any RHS, satisfying the CNF constraint.
2. **Eliminate $\varepsilon$-productions.** Find all *nullable* variables: those that can derive $\varepsilon$ in one or more steps. For every rule $A \to \ldots B \ldots$ where $B$ is nullable, add a copy of the rule with $B$ omitted. Then remove all $A \to \varepsilon$ rules (except $S_0 \to \varepsilon$ if $\varepsilon \in \mathcal{L}(G)$).
3. **Eliminate unit productions.** A *unit production* is $A \to B$ where $B$ is a single variable. For each such rule, add $A \to \alpha$ for every $B \to \alpha$ (non-unit rule). Remove all $A \to B$ rules. Repeat until none remain.
4. **Binarize long rules.** A rule with $k \geq 3$ symbols on the RHS violates CNF. Replace $A \to X_1 X_2 \ldots X_k$ with a chain: $A \to X_1 A_1, \quad A_1 \to X_2 A_2, \quad \ldots, \quad A_{k-2} \to X_{k-1} X_k$ Each $A_i$ is a fresh variable introduced for this rule.
5. **Isolate terminals in mixed rules.** For each terminal $a$ appearing in a rule body of length $\geq 2$, introduce a fresh variable $U_a$ with production $U_a \to a$, and replace every occurrence of $a$ in rule bodies by $U_a$.

# CNF Conversion Algorithm [2]

Steps 1–5 are applied in order; each preserves the language. The result is a CNF grammar for the same CFL. The grammar may be larger, but structurally uniform.

# CNF Conversion Walkthrough

*Example (Convert $G$: $S \to aSb \mid ab$ to CNF.)*:

**Step 1** (new start): no $S$ on any RHS already, so skip.

**Step 2** ($\varepsilon$): neither $S$ rule derives $\varepsilon$ directly. Since $S$ is not nullable, skip.

**Step 3** (unit productions): no unit rules. Skip.

**Step 4** (binarize): $S \to aSb$ has length 3. Introduce $A_1$:

$$S \to aA_1, \quad A_1 \to Sb$$

Rule $S \to ab$ has length 2 (fine as-is, for now).

**Step 5** (isolate terminals): in $S \to aA_1$, $a$ is a terminal in a two-symbol body. Introduce $U_a \to a$ and $U_b \to b$:

$$S \to U_aA_1 \mid U_aU_b, \quad A_1 \to SU_b, \quad U_a \to a, \quad U_b \to b$$

This is the final CNF grammar. Check: every production is $A \to BC$ or $A \to a$. ✓

# Pushdown Automata

**Definition 36**: A *Pushdown Automaton* (PDA) is a 7-tuple $\mathcal{P} = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ where:

- $Q$ is a finite set of states,
- $\Sigma$ is the *input alphabet*,
- $\Gamma$ is the *stack alphabet*,
- $\delta : Q \times (\Sigma \cup \{\varepsilon\}) \times \Gamma \to \mathcal{P}(Q \times \Gamma^*)$ is the *transition function*,
- $q_0 \in Q$ is the *start state*,
- $Z_0 \in \Gamma$ is the *initial stack symbol* (bottom-of-stack marker),
- $F \subseteq Q$ is the set of *accepting states*.

A PDA is an NFA augmented with an unbounded *stack* that it can push to and pop from.

# Pushdown Automata [2]

**Note**: Each transition reads:

- the current *state*,
- the current *input symbol* (or $\varepsilon$ — a "free move"),
- the *top of the stack*,

and produces: a new *state* and a string to *replace* the top stack symbol with. A "pop" is modeled by replacing the top symbol with $\varepsilon$ (nothing).

*Example*: A PDA for $L = \{0^n 1^n \mid n \geq 0\}$:

- On reading 0: push a marker \$ onto the stack.
- On reading the first 1: switch to "popping" mode.
- On reading 1 in popping mode: pop a \$.
- Accept when the input is exhausted and the stack contains only $Z_0$.

# PDA Computation Model

A *configuration* $(q, w, \gamma)$ captures the complete machine state: $q \in Q$ is the current state, $w \in \Sigma^*$ is remaining input, $\gamma \in \Gamma^*$ is the stack (top on left).

**Definition 37**: The *step relation* of a PDA:

$$(q, aw, Z\gamma) \vdash (p, w, \alpha\gamma) \quad \text{iff} \quad (p, \alpha) \in \delta(q, a, Z)$$

Epsilon moves: $(q, w, Z\gamma) \vdash (p, w, \alpha\gamma)$ iff $(p, \alpha) \in \delta(q, \varepsilon, Z)$.

$\vdash^*$ is the reflexive-transitive closure (zero or more steps).

**Definition 38**: Two equivalent *acceptance modes* (both define the same class — CFL):

- **By final state:** $\mathcal{L}(\mathcal{P}) = \{w \mid (q_0, w, Z_0) \vdash^* (f, \varepsilon, \gamma), f \in F\}$
- **By empty stack:** $\mathcal{L}(\mathcal{P}) = \{w \mid (q_0, w, Z_0) \vdash^* (q, \varepsilon, \varepsilon)\}$

**Note**: PDAs are *non-deterministic* by definition. *Deterministic PDAs* (DPDAs) recognize a strict subclass of CFLs. Example: palindromes require non-determinism — a DPDA cannot recognize them.

# PDA Computation Model [2]

LL and LR parsing is a *deterministic* simulation of the PDA for a specific grammar. Parser generators (yacc, ANTLR, Bison) automate exactly this construction.

# PDA Trace for $0^n1^n$

We trace the PDA for $L = \{0^n1^n \mid n \geq 0\}$ on input `0011`:

| Step | State | Input remaining | Stack (top ←) | Action |
|---|---|---|---|---|
| 0 | $q_0$ | `0011` | $Z_0$ | Initial configuration |
| 1 | $q_0$ | `011` | $\$Z_0$ | Read `0`, push \$ |
| 2 | $q_0$ | `11` | $\$\$Z_0$ | Read `0`, push \$ |
| 3 | $q_1$ | `1` | $\$Z_0$ | Read `1`, pop \$ (switch to pop mode) |
| 4 | $q_1$ | | $Z_0$ | Read `1`, pop \$ |
| 5 | $q_{\text{acc}}$ | | $Z_0$ | $\varepsilon$-move, **ACCEPT** |

**Why this works:** each `0` pushes a marker; each `1` pops one. If the counts match, the stack returns to exactly $Z_0$ when input is consumed. If there are more `0`s than `1`s, excess \$ remain. If more `1`s, the stack underflows (no \$ to pop $\Rightarrow$ reject).

# PDA Trace for $0^n1^n$ [2]

**Note**: The PDA non-deterministically guesses **when** to switch from pushing to popping. For $0^n1^n$, a deterministic PDA also works: switch when the first 1 appears.

# The Context-Free Hierarchy

**Theorem 24**: A language is context-free iff it is recognized by some pushdown automaton.

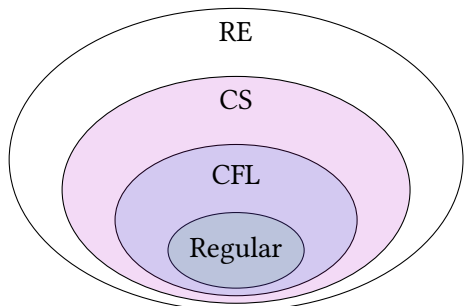
Context-free languages are *strictly more powerful* than regular languages:

- Every regular language is context-free: a DFA is a PDA that ignores its stack entirely.
- $\{a^n b^n \mid n \geq 0\}$ is context-free (by PDA above) but *not* regular (by pumping lemma).

But CFLs still have limits:

- $\{a^n b^n c^n \mid n \geq 0\}$ is **not** context-free.
- Context-sensitive languages (linear-bounded automata) lie above CFLs.

The Chomsky hierarchy captures this nested structure precisely.




**Big picture:** Regular $\subset$ Context-Free $\subset$ Context-Sensitive $\subset$ Recursive (Decidable) $\subset$ RE.

Each level strictly contains the one below, with concrete witness languages separating them.

# CFG $\Leftrightarrow$ PDA Equivalence

**Theorem 25**: $\mathcal{L}$ is context-free iff $\mathcal{L} = \mathcal{L}(\mathcal{P})$ for some PDA $\mathcal{P}$.

**Proof** *(CFG $\Rightarrow$ PDA)*: Given CFG $G$, construct a PDA simulating *leftmost derivation*:

1. *Push* the start symbol $S$ onto the stack.
2. While the stack is non-empty:
   - If top is a *variable* $A$: non-deterministically choose a production $A \to \alpha$ and replace the top with $\alpha$ ($\varepsilon$-move).
   - If top is a *terminal* $a$: read $a$ from input and pop the top.
3. Accept when the stack and input are both empty.

This PDA non-deterministically simulates all leftmost derivations of $G$. □

The forward direction (CFG $\Rightarrow$ PDA) is used by LL/LR parsing: the parser *simulates* the leftmost derivation non-deterministically (with look-ahead to resolve choices). GLL and GLR parsers handle the non-determinism explicitly.

## CFG $\Leftrightarrow$ PDA Equivalence [2]

**Proof** *(PDA $\Rightarrow$ CFG)*: Given PDA $\mathcal{P}$, construct a CFG where each variable $[pXq]$ represents: "$\mathcal{P}$ can go from state $p$ with $X$ on top to state $q$, emptying $X$". The construction is mechanical but tedious; the key insight is that each PDA transition translates into one or two grammar productions. $\square$

# Pumping Lemma for Context-Free Languages

The CFL counterpart of the Pumping Lemma for regular languages, proved using CNF trees.

**Theorem 26** (Pumping Lemma for CFLs (Bar-Hillel, Perles, Shamir, 1961)): Let $L$ be a context-free language. Then there exists $n \geq 1$ such that for every $w \in L$ with $|w| \geq n$, there exist strings $u, v, x, y, z$ with $w = uvxyz$ satisfying:

1. $|vy| \geq 1$ (the "pump" is non-empty),
2. $|vxy| \leq n$ (the pump is not too long),
3. $uv^i xy^i z \in L$ for all $i \geq 0$ (pumping keeps the string in $L$).

Compared to the regular pumping lemma ($w = xyz$, pump $y$), the CFL version pumps *two* substrings $v$ and $y$ *simultaneously*. This reflects the stack: each time the outer variable expands, it contributes one segment $v$ on the left and one segment $y$ on the right.

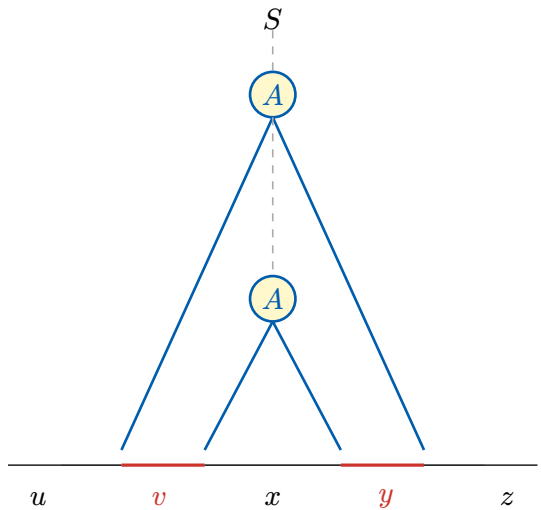
*Example*: Grammar $S \to 0S1 \mid \varepsilon$ generates $\{0^n 1^n\}$. Taking $w = 0^n 1^n$: a valid split is $u = \varepsilon$, $v = 0^k$, $x = \varepsilon$, $y = 1^k$, $z = 0^{n-k} 1^{n-k}$. Then $uv^i xy^i z = 0^{n-k+ik} 1^{n-k+ik} \in L$ for all $i \geq 0$. ✓

# Proof Idea for the CFL Pumping Lemma

Use CNF to make every parse tree binary.
The pumping length is $n = 2^{|V|+1}$, where $|V|$ is the number of variables.

If $|w| \geq 2^{|V|+1}$, the parse tree has height $\geq |V| + 1$.
By the *Pigeonhole Principle*, some root-to-leaf path visits the same variable $A$ *twice*.

1. Let $A_{\text{outer}}$ be the higher occurrence, $A_{\text{inner}}$ the lower.
2. The subtree rooted at $A_{\text{outer}}$ derives $vxy$; the smaller subtree at $A_{\text{inner}}$ derives $x$.
3. $v$ and $y$ are the "side contributions" between the two $A$s.




**Pumping:**

- $i = 0$: replace $A_{\text{outer}}$'s subtree with $A_{\text{inner}}$'s subtree. Result: $uxz \in L$.
- $i \geq 2$: nest another copy of $A_{\text{outer}}$'s subtree inside itself. Result: $uv^ixy^iz \in L$.

# $a^n b^n c^n$ is Not Context-Free

**Theorem 27**: $L = \{a^n b^n c^n \mid n \geq 0\}$ is not context-free.

**Proof**: Assume $L$ is CFL and let $n$ be the pumping length. Choose $w = a^n b^n c^n \in L$ with $|w| = 3n \geq n$.

By the CFL pumping lemma, $w = uvxyz$ with $|vy| \geq 1$ and $|vxy| \leq n$.

Since $|vxy| \leq n < 3n$, the segment $vxy$ can span *at most two* of the three symbol blocks ($a$s, $b$s, $c$s).

- **Case 1:** $v$ and $y$ lie entirely within two adjacent blocks (say $a$s and $b$s).
  Pumping with $i = 2$ increases the count of $a$s and/or $b$s but *not* $c$s.
  The result $uv^2xy^2z$ has unequal symbol counts, so it $\notin L$.
- **Case 2:** $v$ or $y$ spans a block boundary, so pumped strings mix symbols out of order.
  For example, if $v = a^j b^k$, then $v^2 = a^j b^k a^j b^k$, which is not of the form $a^* b^* c^*$. Hence $uv^2xy^2z \notin L$.

In both cases we reach a contradiction. Thus $L$ is not context-free. □

**Intuition:** A stack can maintain *one* counter (*e.g.*, match $a$s with $b$s). Matching $a$s, $b$s, and $c$s *simultaneously* requires *two* independent counters — beyond what any stack can do.

# $ww$ is Not Context-Free

**Theorem 28**: $L = \{ww \mid w \in \{0,1\}^*\}$ is not context-free.

**Proof**: Assume $L$ is CFL with pumping length $n$. Choose $w_0 = 0^n1^n0^n1^n \in L$ (set $w = 0^n1^n$, so $ww = w_0$, and $|w_0| = 4n$).

By the CFL pumping lemma, $w_0 = uvxyz$ with $|vxy| \leq n$. Since $|vxy| \leq n$ and $|w_0| = 4n$, the segment $vxy$ covers at most $n$ consecutive characters of $w_0$. It cannot straddle both the first half $0^n1^n$ and the second half $0^n1^n$ simultaneously.

Hence $v$ and $y$ together modify the symbol counts in *at most one half* of $w_0$.

Pumping with $i = 2$ increases character counts in one half but not the other. The resulting $uv^2xy^2z$ cannot equal $w'w'$ for any $w'$ (the two halves become unequal). Thus $uv^2xy^2z \notin L$, contradicting the lemma. □

**Context:** The language $\{ww \mid w \in \{0,1\}^*\}$ is decidable by Turing machine (it just has two heads, or uses the marking trick from the TM trace slides). This shows Decidable $\neg \subseteq$ CFL.

# Closure Properties of CFLs

| Operation | CFLs closed? | Proof idea |
|---|---|---|
| Union $L_1 \cup L_2$ | ✓ | New start $S \to S_1 \mid S_2$ |
| Concatenation $L_1 \cdot L_2$ | ✓ | New start $S \to S_1 S_2$ |
| Kleene star $L^*$ | ✓ | New start $S \to SS' \mid \varepsilon$ |
| Reversal $L^R$ | ✓ | Reverse all production RHSs |
| Homomorphism $h(L)$ | ✓ | Apply $h$ to each terminal in grammar |
| $\cap$ with regular, $L \cap R$ | ✓ | Product of PDA stack $\times$ DFA state |
| Intersection $L_1 \cap L_2$ | ✗ | $\{a^n b^n\} \cap \{b^n c^n\} = \{a^n b^n c^n\} \notin$ CFL |
| Complement $\overline{L}$ | ✗ | From non-closure under $\cap$ via De Morgan |
| Difference $L_1 \setminus L_2$ | ✗ | $L \setminus R \in$ CFL for regular $R$; but $L_1 \setminus L_2$ may not be |

# Closure Properties of CFLs [2]

**Key contrast with regular languages:** CFLs are NOT closed under intersection or complement. This means there is no direct "product PDA" construction analogous to the DFA product.

Exception: *Deterministic* CFLs (DCFL) **are** closed under complement. DCFL is the language class of most real programming languages (LR(1) grammars).

# Decision Properties of CFLs

***Decidable* questions about CFLs:**

- **Empty?** Is $\mathcal{L}(G) = \emptyset$?
  Check whether $S$ can derive any terminal string.
  $\mathcal{O}(|G|)$ time. ✓
- **Membership:** Is $w \in \mathcal{L}(G)$?
  Use CYK algorithm (CNF required):
  $\mathcal{O}(|w|^3 \cdot |G|)$ time. ✓
- **Finite?** Is $\mathcal{L}(G)$ finite?
  Decidable (check for useless variables).
  $\mathcal{O}(|G|)$ time. ✓

***Undecidable* questions about CFLs:**

- **Universal?** Is $\mathcal{L}(G) = \Sigma^*$?
  *Undecidable*. ✗
- **Intersection empty?** Is $\mathcal{L}(G_1) \cap \mathcal{L}(G_2) = \emptyset$?
  *Undecidable*. ✗
- **Equivalence:** Is $\mathcal{L}(G_1) = \mathcal{L}(G_2)$?
  *Undecidable*. ✗
- **Ambiguous?** Is $G$ ambiguous?
  *Undecidable*. ✗

**The jump from decidable to undecidable happens exactly at intersection and universality** — the same operations that break closure. Decidability and closure are deeply linked.

# Summary of Context-Free Languages

**CFG = PDA**

- CFGs *generate* context-free languages
- PDAs *recognize* context-free languages

**Main Tools**

- Chomsky Normal Form (CNF)
- CYK decides membership in $\mathcal{O}(n^3 \cdot |G|)$
- Parse trees make the structure explicit

**Decidable:** membership, emptiness, and finiteness are decidable.
**Undecidable:** equivalence, universality, ambiguity, and intersection emptiness.

**Closure properties**

- Closed under union, concatenation, star, reversal, homomorphism
- Closed under intersection with a regular language
- **Not** closed under general intersection or complement

**The Limitations**

- One stack handles nested structure.
- One stack does not handle several independent counters at once.

Typical non-CFLs: $\{a^n b^n c^n\}$, $\{ww\}$, $\{w\#w\}$.

# Turing Machines

*"We may compare a man in the process of computing ...to a machine."*

***— Alan Turing, 1936***


Alan Turing


Alonzo Church


Emil Post


Kurt Gödel


Stephen Cook


Richard Karp

# From Automata to General Computation

Finite automata recognize regular patterns, and pushdown automata capture recursive structure. But many computational tasks — arithmetic on unbounded integers, execution of programs, symbolic reasoning, formal verification — require a machine that can *store*, *rewrite*, and *revisit* intermediate results.

A Turing machine is the simplest classical model with exactly this level of generality. It is strong enough to express arbitrary algorithms, yet simple enough to study mathematically.

This section formalizes three central notions: *algorithm*, *decidable problem*, and *computational limit*.

At this point, automata theory becomes the general theory of computation.

# Definition of a Turing Machine

**Definition 39**: A *Turing Machine* (TM) is a 7-tuple $\mathcal{M} = \left(Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}}\right)$ where:

- $Q$ is a finite set of states,
- $\Sigma$ is the *input alphabet* (not containing the blank symbol $\square$),
- $\Gamma$ is the *tape alphabet*, where $\square \in \Gamma$ and $\Sigma \subset \Gamma$,
- $\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$ is the *transition function*,
- $q_0 \in Q$ is the *start state*,
- $q_{\text{accept}} \in Q$ is the *accept state*,
- $q_{\text{reject}} \in Q$ is the *reject state* ($q_{\text{reject}} \neq q_{\text{accept}}$).

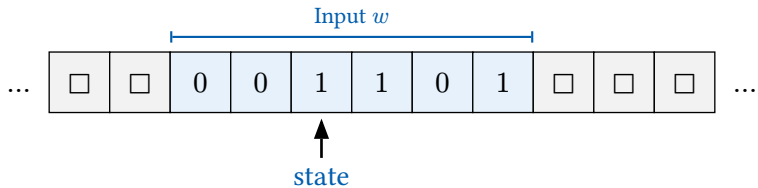
**Note**: Conceptually, a Turing machine has two parts:

- a small *finite control* (the current state), and
- an unbounded *external memory* (the tape).

Compared with a finite automaton, it can move in both directions, rewrite symbols, and halt only when it reaches $q_{\text{accept}}$ or $q_{\text{reject}}$.

# Tape, Head, and State

A Turing machine operates on an infinite tape divided into cells, each containing a symbol from $\Gamma$.



# One Transition Step

The machine starts with the input $w$ written on the tape, the head on the leftmost symbol, in state $q_0$.

At each step:

1. Read the symbol currently under the head.
2. Use the transition function to decide three things:
   - which *state* to enter next,
   - which *symbol* to write in the current cell,
   - whether to move the head *left* or *right*.

For example, if $\delta(q, 0) = (r, \times, R)$, then in state $q$ on symbol $0$ the machine writes $\times$, changes to state $r$, and moves one cell to the right.

# Configurations

**Definition 40**: A *configuration* of a TM is a complete snapshot of the computation. It records the current state, the tape contents, and the head position.

We write configurations as $uqv$. Here $q$ is the current state. String $u$ is the tape content to the left of the head. String $v$ begins with the symbol currently under the head.

**Note**: Once a configuration is fixed, the entire future computation is determined for a deterministic TM.

# Acceptance, Rejection, and Looping

**Definition 41**: A TM $\mathcal{M}$ on input $w$:

- *accepts* $w$ if the computation eventually reaches $q_{\text{accept}}$;
- *rejects* $w$ if it eventually reaches $q_{\text{reject}}$;
- *loops* if it never halts.

**Important distinction:** failing to accept is not the same as rejecting.

A machine may halt and reject. Or it may continue forever without producing any answer. This distinction is central in computability theory.

# Languages Recognized by Turing Machines

**Definition 42**: The language *recognized* by $\mathcal{M}$ is

$$\mathcal{L}(\mathcal{M}) = \{w \in \Sigma^* \mid \mathcal{M} \text{ accepts } w\}.$$

**Note**: If the machine halts on every input, it is a *decider*. Otherwise it is only a *recognizer*.

# A Turing Machine for $0^n1^n$

*Example*: A TM that recognizes $L = \{0^n1^n \mid n \geq 0\}$:

**Operational idea:** match the symbols in pairs.

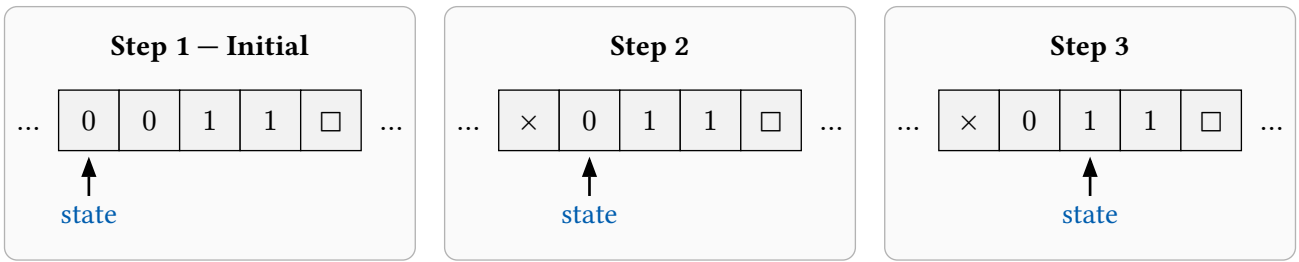
1. Start at the left end and find the first uncrossed 0.
2. Replace it by $\times$ to mark that this 0 has been used.
3. Move right until you encounter the first uncrossed 1.
4. Replace it by $\times$ as well.
5. Return to the left and repeat the same procedure.

The machine accepts when every 0 has been matched with a later 1. It rejects as soon as this matching process breaks down — for example, if a needed 1 does not exist, or if the order is wrong.

**Why the procedure works:** each pass removes exactly one pair $(0, 1)$. Successful completion therefore guarantees both *equal counts* and the correct *left-to-right order*.

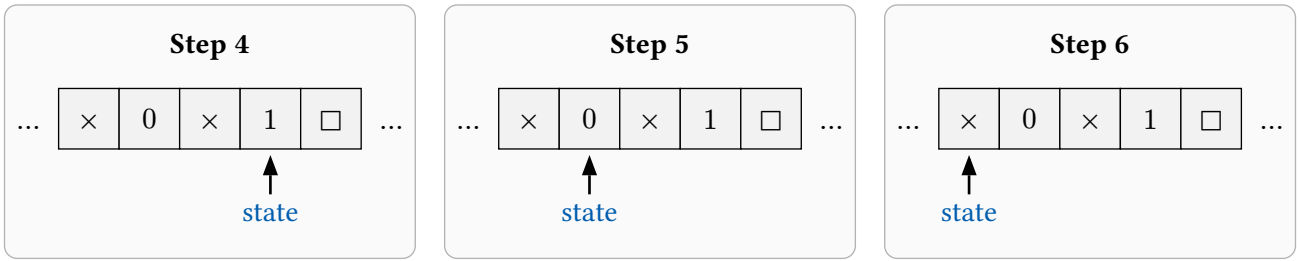
# Trace on 0011 — First Pass

The trace makes the strategy visible: the machine marks one matching 0-1 pair, returns to the left, and repeats the process.



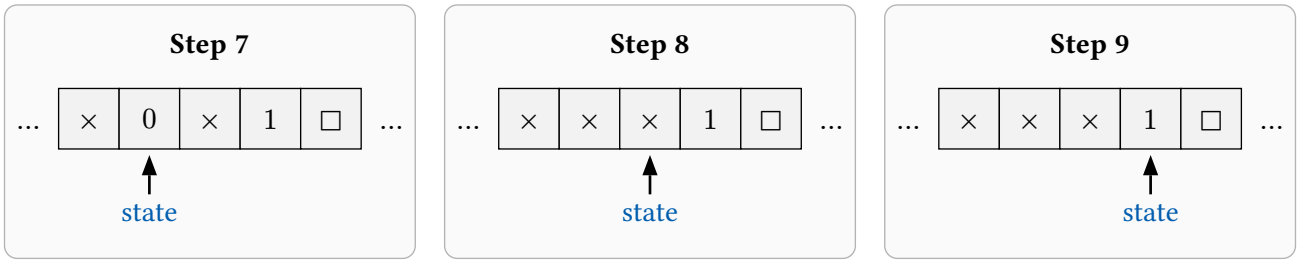

**Note**: The first 0 has now been marked, and the machine is positioned at the 1 that will be paired with it.

# Trace on 0011 — First Pass Continued



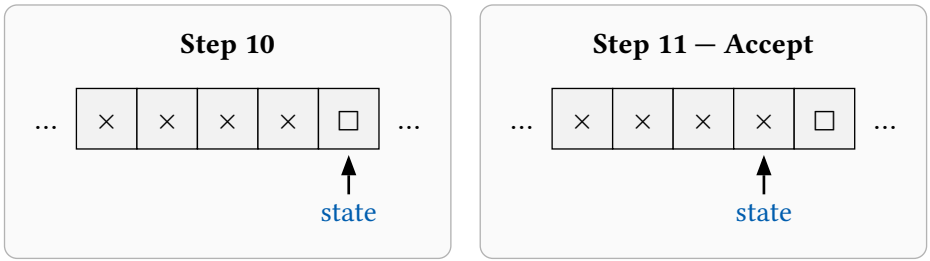

**Note**: One matching pair has been removed. The head has returned to the left end of the relevant tape segment. The machine is ready for the next iteration.

# Trace on 0011 — Second Pass




**Note**: The computation proceeds in the same way on the remaining unmatched symbols.

# Trace on 0011 — Acceptance




Formal trace (abbreviated):

$$q_0\,0011 \Rightarrow \times\, q_1\,011 \Rightarrow \cdots \Rightarrow \times\, q_0\,0 \times 1 \Rightarrow \cdots \Rightarrow \text{accept}$$

**Visual insight:** the tape acts as writable memory. Crossing symbols out lets the machine record its progress without storing large counters in the finite control.

# DFA, PDA, and TM Compared

| Feature | DFA | PDA | TM |
|---|---|---|---|
| Memory | Finite control only | One stack | Read/write tape |
| Input access | Single left-to-right pass | Single pass + stack access | Move left or right |
| Can modify memory? | No | Only push/pop | Yes |
| Typical task | Pattern matching | Parsing nested syntax | General algorithmic computation |
| Halting | Always halts | Always halts | May run forever |
| Language class | Regular | Context-free | Decidable / recognizable |

**Historical context:** Alan Turing introduced his machine model in 1936, before electronic computers existed. He was trying to formalize the notion of "mechanical procedure" to resolve Hilbert's *Entscheidungsproblem* (decision problem). The Turing machine remains the standard reference model of computation.

# Variants of Turing Machines

Several variants of TMs exist, all *equivalent in computational power*:

- *Multi-tape TMs:* several tapes with independent heads.
- *Non-deterministic TMs:* $\delta$ may branch into several possible moves.
- *Two-way infinite tape:* the head may roam freely in both directions.
- *Multi-track or multi-head TMs:* more structured storage, but the same computability.

**Theorem 29**: Every multi-tape TM can be simulated by a single-tape TM.

**Proof**: Encode all virtual tapes on one large tape using separators and marked head positions. One simulation round scans this encoding, updates the marked cells, and writes the new symbols. □

**Theorem 30**: Every non-deterministic TM can be simulated by a deterministic TM.

**Proof**: Explore the entire computation tree systematically, for example breadth-first. If some branch accepts, the deterministic simulator will eventually discover it. □

# Variants of Turing Machines [2]

**Note**: These simulations may slow computation down — sometimes polynomially, sometimes exponentially — but they do *not* change which problems are computable.

The equivalence of deterministic and non-deterministic TMs is *not* about efficiency.

Whether they can solve the same problems *efficiently* is the famous **P vs NP** problem — one of the most important open questions in mathematics and computer science.

# Equivalent Models of Computation

Turing machines are not the only formal model of algorithmic computation. What makes the notion of computability convincing is its *robustness*: several independently invented formalisms turn out to define exactly the same class of computable functions.

| Model | Main idea | Where it appears |
|---|---|---|
| Lambda calculus | Computation by substitution | Logic, proof theory, functional programming |
| Recursive functions | Composition, primitive recursion, minimization | Number theory and mathematical logic |
| Register / RAM machines | Instructions on integer registers | Assembly language and algorithm design |
| Boolean circuits | Finite networks of gates | Hardware and complexity theory |
| Cellular automata | Simple local update rules | Physics, emergence, simulation |

**Key insight:** these models arise from logic, arithmetic, hardware, and programming — yet they all capture the same notion of an *effective procedure*.

# Church–Turing Thesis

**Thesis 31**: Every "effectively computable" function is computable by a Turing machine.

This is a *thesis*, not a theorem — it cannot be proved mathematically because "effectively computable" is an informal notion. However, every formal model of computation ever proposed has turned out to be *equivalent* to Turing machines:

- Lambda calculus (Church, 1936)
- Recursive functions (Gödel, Kleene)
- Post systems (Post, 1936)
- Register machines
- Modern programming languages (C, Python, Haskell, ...)

The Church–Turing thesis says that Turing machines capture *everything* that can be computed.

If a problem cannot be solved by a TM, it cannot be solved by *any* mechanical procedure — regardless of the programming language or hardware.

# Decidability and Recognizability

**Definition 43**: A language $L$ is *Turing-recognizable* — also called *semi-decidable* or *recursively enumerable* (**RE**) — if some TM accepts every $w \in L$. On inputs $w \notin L$, the machine may reject or loop forever.

**Definition 44**: A language $L$ is *decidable* — also called *recursive* (**R**) — if some TM halts on *every* input and always gives the correct yes/no answer.

**Definition 45**: A language $L$ is *co-recognizable* (**co-RE**) if its complement $\overline{L}$ is Turing-recognizable.

**Operational distinction:** a *decider* always halts with a correct yes/no answer. A *recognizer* is only required to halt on yes-instances.

# Decidability and Recognizability [2]

**Theorem 32**: A language $L$ is decidable iff $L \in \mathrm{RE}$ and $L \in \text{co-RE}$, *i.e.*, $\mathrm{R} = \mathrm{RE} \cap \text{co-RE}$.

**Proof**: **(⇒)** Assume $L$ is decidable. Let $M$ be a decider for $L$.

- $M$ recognizes $L$, so $L \in \mathrm{RE}$.
- Swapping the accept and reject states of $M$ gives a decider for $\overline{L}$, so $L \in \text{co-RE}$.

**(⇐)** Now assume $L \in \mathrm{RE}$ and $L \in \text{co-RE}$.

Let $M_{\text{yes}}$ recognize $L$ and $M_{\text{no}}$ recognize $\overline{L}$.

We construct a decider using the *dovetailing technique*:

- On input $w$, alternate: one step of $M_{\text{yes}}$, then one step of $M_{\text{no}}$, and so on.
- Since $w \in L$ or $w \in \overline{L}$, one machine eventually accepts.
- If $M_{\text{yes}}$ accepts, accept. If $M_{\text{no}}$ accepts, reject.

Dovetailing ensures neither simulation is starved, so the decider always halts. □

# Examples of Computability Classes

| Problem | Class | Reason |
|---|---|---|
| DFA membership | D | Run the automaton once; it always halts |
| CFG membership | D | Use CYK or another parsing algorithm |
| SAT | D | There are finitely many assignments to try |
| HALT | RE $\setminus$ R | Simulate and accept when halting is observed |
| $\overline{\text{HALT}}$ | co-RE $\setminus$ R | Its complement is HALT |
| $\text{REGULAR}_{\text{TM}}$ | Neither | No recognizer exists in either direction |

*Semi-decidable* problems often arise when a positive witness can be found by search, but a negative answer would require proving that *no witness exists*.

# Programs as Data

To ask "does $M$ halt on $w$?" we need to *feed $M$ itself as an input* to another TM. That requires encoding machines as strings.

**Definition 46**: The *encoding* $\langle M \rangle \in \{0,1\}^*$ of a TM $M$ is a canonical binary string representation of its description: the state list, alphabet, and transition table, encoded in some fixed format.

We write $\langle M, w \rangle$ for the encoding of the pair $(M, w)$.

**Key insight:** TMs can read their *own descriptions* as input. This is *programs as data* — and it is why the Halting Problem is well-formed. The question "does $M$ halt on $\langle M \rangle$?" is mathematically well-defined.

# Programs as Data [2]

**Theorem 33** (Counting Theorem): There exist languages over $\{0, 1\}$ that are not Turing-recognizable.

**Proof**: There are only *countably many* TMs, because each machine description is a finite string. But there are *uncountably many* languages, since $\mathcal{P}(\{0, 1\}^*)$ is uncountable by Cantor's theorem. So most languages cannot be matched with any TM at all. □

**This perspective underlies modern computing:**

- *Compilers:* programs that take source code as input.
- *Interpreters:* programs that simulate another program on data.
- *Formal verification:* tools that analyse program behaviour — and therefore encounter exactly the limits exposed by undecidability results.

# Universal Machines

**Theorem 34**: There exists a single TM $U$ such that, on input $\langle M, w \rangle$, the machine $U$ simulates the computation of $M$ on $w$.

**Proof**: The encoded transition table of $M$ is part of the input. Machine $U$ stores a simulated configuration of $M$ on its own tape and repeatedly applies the next transition described by that table. So one fixed machine can imitate any other machine. □

**Why this matters:** interpreters, emulators, virtual machines, and stored-program computers all rely on this idea. A program is just *data that another program can execute*.

# Many-One Reductions

**Definition 47**: A *many-one reduction* from $A$ to $B$, written $A \leq_m B$, is a computable function $f$ such that

$$x \in A \iff f(x) \in B$$

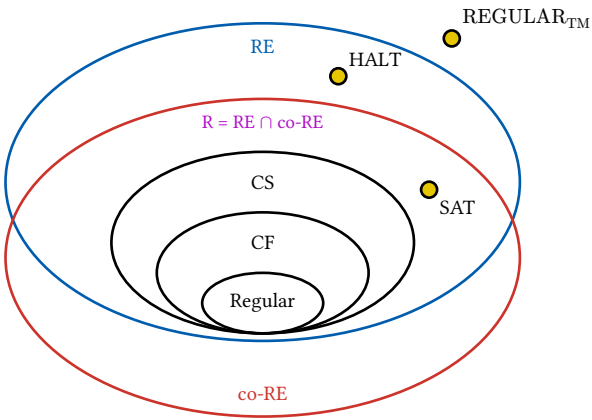
for every input $x$.

**Theorem 35**: If $A \leq_m B$ and $B$ is decidable, then $A$ is decidable.

**Proof**: To decide whether $x \in A$, compute $f(x)$ and run the decider for $B$ on the transformed input $f(x)$. □

To prove that $B$ is undecidable, it is enough to reduce a known undecidable problem $A$ to it. Reductions let us *transfer impossibility* from one problem to another.

# Map of Decidability and Recognizability




**Classes overview:**

$$\text{Regular} \subset \text{CF} \subset \text{CS} \subset \text{R} \subset \text{RE}$$

- **R** = **RE** ∩ **co-RE** — decidable languages
- **HALT** and $\mathbf{A}_{\mathbf{TM}}$ lie in **RE** \ **R** — yes-instances can be confirmed by simulation
- $\overline{\text{HALT}}$ lies in **co-RE** \ **R**
- **SAT** lies in **R** — decidable by exhaustive search; NP-complete
- $\mathbf{REGULAR}_{\mathbf{TM}}$ lies in neither **RE** nor **co-RE**

**Warning:** Decidable ⊊ Recognizable.

- There exist languages that are recognizable but *not* decidable (*e.g.*, HALT).
- Some languages are *not recognizable* (for example $\overline{\text{HALT}}$).
- Some are in neither **RE** nor **co-RE** (for example $\text{REGULAR}_{\text{TM}}$).

# The Halting Problem

**Definition 48** (Halting Problem): Given a TM $\mathcal{M}$ and an input $w$, does $\mathcal{M}$ halt on $w$?

$$\text{HALT} = \{\langle \mathcal{M}, w \rangle \mid \mathcal{M} \text{ is a TM that halts on input } w\}$$

**Theorem 36**: HALT is *undecidable*.

**Proof**: By *diagonalization*. Assume for contradiction that some TM $H$ decides HALT.

Construct a TM $D$ that, on input $\langle \mathcal{M} \rangle$:

1. Run $H$ on $\langle \mathcal{M}, \langle \mathcal{M} \rangle \rangle$.
2. If $H$ accepts (*i.e.*, $\mathcal{M}$ halts on $\langle \mathcal{M} \rangle$), then *loop forever*.
3. If $H$ rejects, then *accept*.

Now consider $D$ on input $\langle D \rangle$:

- If $D$ halts on $\langle D \rangle$, then $H$ accepts, so $D$ loops. Contradiction.
- If $D$ does not halt on $\langle D \rangle$, then $H$ rejects, so $D$ accepts (halts). Contradiction.

# The Halting Problem [2]

In either case we get a contradiction, so $H$ cannot exist. □

**Key insight:** There exist well-defined mathematical problems that *no algorithm can solve*.

This is not a limitation of current technology — it is a *fundamental* impossibility. The proof is a precise self-reference paradox: the machine $D$ is forced to halt iff it does *not* halt.

This connects computation to Cantor's diagonal argument, Gödel's incompleteness ideas, and deep philosophical questions about the limits of mechanical reasoning.

# The Acceptance Problem

**Definition 49** (Acceptance Problem):

$$\mathrm{A}_{\mathrm{TM}} = \{\langle M, w\rangle \mid M \text{ accepts } w\}$$

**Theorem 37**: $\mathrm{A}_{\mathrm{TM}}$ is Turing-recognizable but undecidable.

**Proof**: *Recognizable:* simulate $M$ on $w$. If $M$ accepts, accept. If $M$ rejects or loops, our simulator may simply run forever.

*Undecidable:* reduce HALT to $\mathrm{A}_{\mathrm{TM}}$. Given $\langle M, w\rangle$, construct a machine $N$ that ignores its own input, simulates $M$ on $w$, and accepts iff that simulation ever halts. Then

$$\langle M, w\rangle \in \mathrm{HALT} \iff \langle N, 0\rangle \in \mathrm{A}_{\mathrm{TM}}$$

so a decider for $\mathrm{A}_{\mathrm{TM}}$ would also decide HALT. □

This is the prototypical *semi-decidable* problem: successful computations can be witnessed, but unsuccessful ones may leave us waiting forever.

# Rice's Theorem

The Halting Problem is just one undecidable problem. Rice's theorem shows that *any* non-trivial property of what a TM computes is undecidable.

**Definition 50** (Semantic property): A property $P$ of Turing machines is *semantic* if it depends only on the *language recognized* by the TM, not on its internal structure. Formally: if $\mathcal{L}(M_1) = \mathcal{L}(M_2)$, then $P(M_1) \iff P(M_2)$.

A semantic property is *non-trivial* if some TMs satisfy it and some do not.

*Example*:

- "$\mathcal{L}(M)$ *is finite*" — semantic (about the language), non-trivial.
- "$\mathcal{L}(M)$ *is regular*" — semantic, non-trivial ($\text{REGULAR}_{\text{TM}}$).
- "$\mathcal{L}(M) = \varnothing$" — semantic, non-trivial ($\text{EMPTY}_{\text{TM}}$).
- "$M$ *has at most 5 states*" — **not** semantic (depends on machine, not its language).

# Rice's Theorem [2]

**Theorem 38** (Rice's Theorem): Every non-trivial semantic property of TMs is undecidable. That is, if $P$ is non-trivial and semantic, then $\{\langle M \rangle \mid P(M)\}$ is undecidable.

**Proof**: Assume WLOG that $P(M_\emptyset) = \text{false}$ where $\mathcal{L}(M_\emptyset) = \emptyset$. Since $P$ is non-trivial, there exists some machine $M_P$ with $P(M_P) = \text{true}$.

Reduce HALT to $P$: given $\langle M, w \rangle$, build a machine $M'$ that on input $x$ first simulates $M$ on $w$.

- If the simulation of $M$ on $w$ ever halts, then $M'$ continues by simulating $M_P$ on $x$.
- If the simulation never halts, then $M'$ never reaches the second phase.

Therefore:

- if $M$ halts on $w$, then $\mathcal{L}(M') = \mathcal{L}(M_P)$, so $P(M') = \text{true}$;
- if $M$ does not halt on $w$, then $\mathcal{L}(M') = \emptyset$, so $P(M') = \text{false}$.

A decider for $P$ would therefore decide HALT, which is impossible. □

# Consequences of Rice's Theorem

| Property of $\mathcal{L}(M)$ | Decidable? | Reason |
|---|---|---|
| $\mathcal{L}(M) = \emptyset$ (empty language) | ✗ | Semantic, non-trivial: Rice |
| $\mathcal{L}(M)$ is finite | ✗ | Semantic, non-trivial: Rice |
| $\mathcal{L}(M)$ is infinite | ✗ | Semantic, non-trivial: Rice |
| $\mathcal{L}(M)$ contains string $w_0$ | ✗ | Semantic, non-trivial: Rice |
| $\mathcal{L}(M)$ is regular | ✗ | Semantic, non-trivial: Rice |
| $\mathcal{L}(M) = \Sigma^*$ (accepts all) | ✗ | Semantic, non-trivial: Rice |
| $\mathcal{L}(M_1) = \mathcal{L}(M_2)$ (equivalence) | ✗ | Semantic, non-trivial: Rice |
| **$M$ has fewer than 100 states** | ✓ | **Syntactic** (structural) — NOT semantic! |
| **$M$ halts on the empty string $\varepsilon$ within 100 steps** | ✓ | **Syntactic:** simulate 100 steps directly |
| **Is $w \in \mathcal{L}(M)$?** for a *fixed known decider* $M$ | ✓ | Run $M$ on $w$; by assumption it always halts |

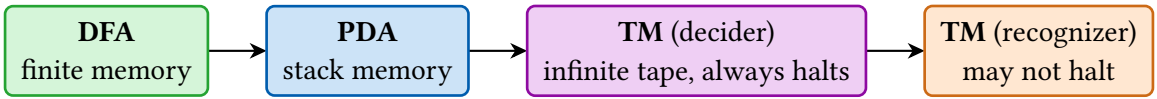
# Consequences of Rice's Theorem [2]

**The critical distinction:** a property is *decidable* only when it depends on the *machine description* (syntactic) rather than on the *language it computes* (semantic).

"How many states does $M$ have?" — we can read off the encoding. Decidable. "Does $M$ accept every input?" — this is about the language. Undecidable.

**Practical consequence:** No static analysis tool, linter, or type checker can ever be *both complete and sound* for general semantic properties of programs. Every real tool is either *approximate* (may miss bugs) or *conservative* (may report false positives).

# The Landscape of Computability




| Language Class | Machine | Closure |
|---|---|---|
| Regular | DFA/NFA | All Boolean operations |
| Context-Free | PDA | Union, concat, star; not intersection or complement |
| Decidable | TM (decider) | All Boolean operations |
| Recognizable | TM (recognizer) | Union, intersection; not complement |

# Exercises on Turing Machines

1. **Design a TM** for $L = \{a^n b^n c^n \mid n \geq 0\}$. Describe the states and tape actions in plain English (no need for a full transition table). What does the tape look like mid-computation?
2. **Tracing:** Run the TM for $0^n 1^n$ (from the trace slides) on input `000111`. Write out the full configuration sequence. What happens on input `0011`? On `001`?
3. **Variants:** A two-tape TM can copy the first half of its input to the second tape. Use this to sketch a TM for $L = \{ww \mid w \in \{0,1\}^*\}$. Why is this easy with two tapes but hard with one?
4. **Decidability:** Classify each property as decidable (D), recognizable but not decidable (R), or not even recognizable (N). Justify each answer:
   - $\{\langle M \rangle \mid M \text{ accepts the empty string}\}$
   - $\{\langle M \rangle \mid M \text{ halts on all inputs}\}$
   - $\{\langle M \rangle \mid M \text{ has exactly 7 states}\}$
   - $\{\langle M_1, M_2 \rangle \mid \mathcal{L}(M_1) = \mathcal{L}(M_2)\}$
5. **Rice's Theorem:** For each property below, state whether Rice's Theorem applies. If it does, conclude undecidability. If it does not, say why (and determine decidability separately):

# Exercises on Turing Machines [2]

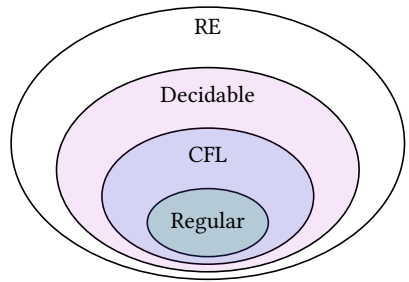
- $\mathcal{L}(M)$ is a regular language.
- $M$ visits state $q_3$ on input $\varepsilon$.
- $M$ accepts at least one palindrome.
- $M$ has a transition on symbol \$.

**Challenge:** prove that $\text{HALT} \leq_m \{\langle M \rangle \mid \mathcal{L}(M) \neq \emptyset\}$ by an explicit reduction. Then conclude $\text{EMPTY}_{\text{TM}}$ is undecidable.

# Summary of Formal Languages and Automata

1. **Formal languages:** every decision problem can be encoded as a language $L \subseteq \Sigma^*$.
   Deciding the problem means deciding whether the input word belongs to $L$.

2. **Regular languages** (DFA = NFA = $\varepsilon$-NFA = RegExp) — Kleene's Theorem.
   - Memory model: finitely many states.
   - Pumping Lemma: a *necessary* test for non-regularity.
   - Myhill-Nerode: a *necessary and sufficient* characterization.
   - Closure and decision properties are exceptionally strong.




3. **Context-free languages** (CFG = PDA):
   - Memory model: one unbounded stack.
   - CNF and CYK provide a uniform recognition method in $\mathcal{O}(n^3)$ time.
   - CFL Pumping Lemma reveals a higher expressiveness than regular languages.
   - Important questions are still undecidable, including equivalence and ambiguity.

# Summary of Turing Machines and Computational Limits

1. **Turing machines** give a mathematically precise model of general computation.
   - They have finite control together with unbounded read/write memory.
   - Many alternative models are equivalent in expressive power.
2. **Decidability landscape:**
   - **R** consists of the decidable languages.
   - **RE** consists of the recognizable languages.
   - Recognizability is strictly weaker than decidability.
   - In fact, $\mathrm{R} \subsetneq \mathrm{RE}$ and $\mathrm{R} = \mathrm{RE} \cap \text{co-RE}$.

The *language hierarchy*:

$$\text{Regular} \subsetneq \text{CFL} \subsetneq \text{Decidable} \subsetneq \text{RE} \subsetneq \text{All Languages}$$

Each step up requires a strictly more powerful machine model:

$$\text{DFA} \prec \text{PDA} \prec \text{TM-decider} \prec \text{TM-recognizer} \prec \text{✨}$$

3. **Undecidability is unavoidable:**
   - **HALT** is undecidable by diagonalization.
   - Reductions transfer undecidability from one problem to another.
   - **Rice's Theorem** rules out decision procedures for non-trivial semantic properties of programs.
4. **Big picture:** computation has extraordinary power, but it also has absolute mathematical limits.
5. **Counts argument:** countably many TMs, uncountably many languages — most are not even RE.